Maximal divergence of the leximin and leximax core selections in exact games

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Abstract

A singleton-normalized exact transferable-utility game carries two canonical egalitarian core selections: the leximin allocation Lmin and the leximax-min allocation Lmax. They coincide on every exact game with at most four players. We measure their separation by $\rho(v) = \|\operatorname{Lmin}(v) - \operatorname{Lmax}(v)\|_1/v(N)$ and let $s(n)$ be its supremum on n-player exact games. We prove $s(5) = 1/2$, $s(6) = 2/3$, and $\sup_n s(n) = 2$. An explicit two-parameter family $G(p,k)$ satisfies

$$\rho(G(p,k)) = \frac{2(p(k-1)-1)}{k(p+1)}, \qquad 2 - \rho(G(p,k)) = \frac{2n}{k(p+1)}, \quad n = p+1+k,$$

and yields $s(n) \geq 2 - 8n/(n^2 - (n \bmod 2))$. A common family of tight caps drives both selections: leximin treats each cap as a ceiling and concentrates the residual on one player, whereas leximax-min treats it as a budget and spreads the residual, making the two supports asymptotically disjoint. On the two-generator class we also give a dual upper-cap description of the core and a one-coalition certificate for the leximin floor; the mirror leximax-min ceiling statement fails.

The fixed-dimensional upper bounds are computer-assisted. At five players, two exact pipelines exhaust all configurations and use independently formulated Lmin-optimality components; they prove the sharp value and the 1/3 bound in the leximin $(4,1)$ regime. At six players, singleton exactness and lexicographic optimality map a strict counterexample into one of fifteen sign-labelled blocker systems. Exact proof objects certify all fifteen infeasible; the largest systems reduce to nineteen role-level exchange schemas and three negative-cut templates. The human counterexample-to-relaxation implication, finite certificates, and novelty determination are kept as separate review obligations.

Keywords: cooperative games; exact games; core; leximin; leximax; Lorenz order; computer-assisted proof.
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1 Introduction

Let (N, v) be a transferable-utility game on $N = \{1, \ldots, n\}$ with

$$\operatorname{Core}(v) = \{x \in \mathbb{R}^N : x(N) = v(N),\ x(S) \geq v(S) \text{ for all } S \subseteq N\},$$

where $x(S) = \sum_{i \in S} x_i$. The game is *exact* if every coalition inequality is attained by some core allocation, and *singleton-normalized* if $v(\{i\}) = 0$ for every i; on that domain every core allocation is nonnegative. Exact games are the natural home for egalitarian core selections, and two such selections are canonical. The *leximin* allocation $\operatorname{Lmin}(v)$ maximizes the increasing rearrangement

of x lexicographically over the core; the *leximax-min* allocation $\mathrm{Lmax}(v)$ minimizes the decreasing rearrangement lexicographically over the core. Both are single-valued on a nonempty compact convex set. The first raises the poorest as far as possible, the second lowers the richest as far as possible.

On small games the two agree. The companion paper (Flanagan, 2026) proves that every singleton-normalized exact game on at most four players is an exact partition game at Lmin; by the established implication from exact partition to Lorenz domination (Llerena and Mauri, 2017), Lmin then Lorenz-dominates every core allocation, and a Lorenz-greatest core point is simultaneously leximin- and leximax-optimal[1], so $\mathrm{Lmin}(v) = \mathrm{Lmax}(v)$ for every singleton-normalized exact game with $n \leq 4$. Our complete enumerations at $n \leq 4$ reconfirm this rigidity as a control (Section 5). The adjacent large-core chain — exact games on at most four players have large cores (Biswas et al., 1999), and on large-core games the procedural egalitarian solution coincides with Lmax (Dietzenbacher, 2021; Arin et al., 2003) — pins only the Lmax side: it does not deliver the coincidence, and indeed a five-player exact game with a large core has Lmin $\neq$ Lmax: the lower envelope of $(2, 0, 12, 6, 0)$, $(0, 0, 0, 0, 20)$ and $(0, 2, 10, 4, 4)$, with $\| \mathrm{Lmin} - \mathrm{Lmax} \|_1 = v(N)/5$ (verified exactly; archived with the repository). The companion's lower-envelope realization technique is used again here.

The pair itself has recent precedent. Vazirani (2024) and Gangam et al. (2024) study leximin and leximax core imputations of assignment, flow and matching games, observe on examples that the two selections differ, and describe their opposition in exactly the direction this paper quantifies; the leximax selection appears earlier as the leximax solution of neighbour games (Klijn et al., 2003) and as the lexmax solution in the egalitarian-set literature (Llerena and Mauri, 2016), besides its large-core characterization (Arin et al., 2003). The candidate contribution here is the quantitative worst case: the exact domain, the supremum 2 with its support-disjointness characterization, the closed-form extremal family, and the direction result that the *concentrating* selection is the leximin one. A targeted review of the principal egalitarian-core, antiduality, exact-partition, large-core, and exact-cone sources found no prior fixed-dimensional separation bound (Arin et al., 2008; Dietzenbacher and Yanovskaya, 2020; Studený and Kratochvíl, 2022). The closest adjacent base-polyhedron theory studies one decreasingly minimal or lexicographically optimal base and its canonical partitions (Frank and Murota, 2022). More sharply, Schoot Uiterkamp (2024) characterizes compact convex sets having a least translated and rescaled majorized point for every translation as base-polyhedra, and derives the corresponding characterization of convex TU games. That theorem concerns a least-majorized point after every strategic translation, not the separation of two opposing selectors in one arbitrary exact-game core. The exact and totally balanced cone papers classify facets in game-value space (Kroupa and Studený, 2019; Studený and Kratochvíl, 2022). This is not a hidden change of vocabulary: if an exact game's core is directly the base-polyhedron $B(f)$ of a submodular f, exactness gives $v(S) = \min_{x \in B(f)} x(S) = f(N) - f(N \setminus S)$, so v is convex. Thus the direct base-polyhedron transfer covers the convex subdomain, not arbitrary exact games. On the special two-bound domain, where the core is a box intersected with the efficiency hyperplane, the egalitarian core is a singleton given by a clipped- level formula (Gong et al., 2022); hence the opposing Lorenz selections coincide there. The later one-bound class is a reduced-game-stable subclass of the same coincidence domain (Gong et al., 2024). Exact-core-bound reductions now extend to all balanced games, but the resulting coordinate-bound equality values can lie outside the core already with three players (Gong and Dietzenbacher, 2026). Thus neither checked formulation

[1]With equal totals, Lorenz domination gives $\sum_{i \leq k} x_i^{\uparrow} \geq \sum_{i \leq k} y_i^{\uparrow}$ for every k; induction on k yields $x^{\uparrow} \geq_{\mathrm{lex}} y^{\uparrow}$, and complementing partial sums gives the leximax-min half. See Dutta and Ray (1989) for the Lorenz-dominant core element on convex games.

states or implies a distance theorem for the two opposing selections on arbitrary exact cores. This negative search is not a novelty proof, and the claim remains subject to specialist review. The Lorenz-monotonic allocation-scheme framework of Izquierdo et al. (2024) is another close terminological neighbor: its allocations are indexed across subgames and its exactness strengthening characterizes convexity, rather than bounding the two selections within one exact core. Likewise, the maximal population-monotonic egalitarian domain of Dietzenbacher and Doğan (2025) characterizes a Dutta–Ray allocation scheme across populations rather than the separation of the two selections in one core.

The question of this paper is what happens after the rigidity breaks. Write

$$\rho(v) = \frac{\|\operatorname{Lmin}(v) - \operatorname{Lmax}(v)\|_1}{v(N)}, \qquad s(n) = \sup\{\rho(v) : \ v \text{ exact on } n \text{ players}, \ v(N) > 0\},$$

the supremum running over singleton-normalized games. Since $\operatorname{Lmin}(v)$ and $\operatorname{Lmax}(v)$ are nonnegative with the same total $v(N)$, we have $\rho \le 2$ always, and $\rho = 2$ would mean disjoint supports. The rigidity says $s(n) = 0$ for $n \le 4$. How large can $s(n)$ get?

Results. The answer is: as large as arithmetically possible.

- $\sup_n s(n) = 2$, and at fixed n, $s(n) \ge 2 - 8n/(n^2 - (n \bmod 2))$, which is $2 - 8/n$ for even n (Theorem 17); in particular $s(n) \ge 1$ from $n = 8$ and $s(n) > 1$ from $n = 9$; the bound is the full exact-class value at $n = 6$, and is attained on the two-generator class at $n = 7, 8$ by complete enumeration.
- The witnesses form a two-parameter family $G(p, k)$ whose two allocations are computed in closed form (Theorem 13), with the deficit identity $2 - \rho(G(p, k)) = 2n/(k(p + 1))$ (Corollary 15).
- On the two-generator class the core has a dual description as a system of upper caps (Theorem 6), from which the leximin floor inherits a one-coalition certificate (Theorem 7); the mirror statement for the leximax-min ceiling is false, and we give the smallest counterexample (Observation 10).
- $n = 5$ is anomalous: the family formula gives $1/3$, a game analysed by hand attains $1/2$ (Theorem 21), and the value is the supremum — $s(5) = 1/2$, with $\rho \le 1/3$ in the $(4, 1)$ leximin regime (Theorems 33 and 32, by configuration-complete finite check, twice over) — while the $n = 5$ extremal class is qualitatively unlike the $n \ge 6$ one.
- At $n = 6$ the family lower bound is sharp: $s(6) = 2/3$ (Theorem 35). The upper bound is a table-free exact finite proof using singleton exactness witnesses and lexicographic exchange blockers; all fifteen sign-count cases have independently checked CPC proof objects.

The mechanism, in one sentence. In $G(p, k)$ a single family of tight caps forces $p + 1$ players to share only p; leximin reads each cap as a ceiling, so the floor sticks at $p/(p + 1)$ and the entire residual piles onto one player, while leximax-min reads the same cap as a budget, loading half the players up to $p(k - 1)/k$ and thereby squeezing the capped players down to $1/k$ — so the two supports become asymptotically disjoint and $\rho \to 2$. It is worth stating the direction explicitly, because it inverts the usual reading: the *concentrating* selection is the leximin one. In $G(p, k)$ the leximin allocation gives one player a share $p(k - 1)/((p + 1)k) \to 1$ of $v(N)$ while everyone else sits just below 1; it is nevertheless leximin-optimal, because nothing better is feasible for the poorest. Both selections are driven into extreme shapes by the same constraint, in opposite directions.

What is not proved. An upper bound matching Theorem 17 at fixed n is open, and Section 6 shows that one natural route to it is closed. Section 7 separates the proved statements from the enumerated and the conjectured ones, and lists the open problems.

2 Preliminaries

Throughout, $N = \{1, \dots, n\}$, all games are singleton-normalized, and $T := v(N) > 0$. Vectors are indexed by N, and $\mathrm{supp}(x) = \{i : x_i \neq 0\}$.

2.1 Exact games as lower envelopes

The following generation fact is the only construction used in this paper. It is standard; see Csóka et al. (2011) and Studený and Kratochvíl (2022) for the surrounding theory of the exact cone, and Flanagan (2026) for the same statement in the companion.

Lemma 1 (lower-envelope generation)**.** *Let $g_1, \dots, g_r \in \mathbb{R}^N_+$ have a common total $g_1(N) = \cdots = g_r(N) = T$, and set $v(S) = \min_j g_j(S)$. Then v is exact, $v(N) = T$, every g_j lies in* Core(v)*, and v is singleton-normalized as soon as every coordinate vanishes in some g_j. Conversely, every exact singleton-normalized game is of this form: one may take $g_1, \dots, g_r$ to be the vertices of* Core(v)*.*

Proof. Each g_j satisfies $g_j(S) \geq v(S)$ by definition of the minimum and $g_j(N) = T = v(N)$, so $g_j \in \mathrm{Core}(v)$; and the inequality at S is attained by whichever g_j realizes the minimum, so v is exact. The singleton statement is immediate from $v(\{i\}) = \min_j (g_j)_i$. Conversely, if v is exact then $v(S) = \min_{x \in \mathrm{Core}(v)} x(S)$ for every S, and the minimum of a linear functional over the polytope Core(v) is attained at a vertex; the vertices are nonnegative because v is singleton-normalized, and all have total T. □

We call a game generated as in Lemma 1 by r allocations an *r-generator* game. Part of what follows is specific to $r = 2$.

2.2 The two selections

For $x \in \mathbb{R}^N_+$ let $x^{\uparrow}$ be its increasing and $x^{\downarrow}$ its decreasing rearrangement.

Definition 2. Lmin(v) *is the allocation in* Core(v) *whose increasing rearrangement is lexicographically largest;* Lmax(v) *is the allocation in* Core(v) *whose decreasing rearrangement is lexicographically smallest.*

Both exist and are unique on a nonempty compact convex set: existence is compactness applied stage by stage, and uniqueness follows by comparing two putative optima with their midpoint, which is feasible by convexity and strictly better at the first stage where they differ. We write $\mu(v) = \min_i \mathrm{Lmin}(v)_i$ for the *leximin floor* and $M(v) = \max_i \mathrm{Lmax}(v)_i$ for the *leximax-min ceiling*; equivalently $\mu = \max_{x \in \mathrm{Core}} \min_i x_i$ and $M = \min_{x \in \mathrm{Core}} \max_i x_i$.

2.3 Normalization and the ratio

Two conventions have to be stated honestly, because neither selection is translation-equivariant: both are applied to the raw payoff vector x, not to $x - (v(\{i\}))_{i \in N}$, and on a game that is not singleton-normalized these are different objects — the leximax-min allocation in particular changes under the shift. We therefore fix the domain once and for all: all games are singleton-normalized,

and both selections are computed on the normalized game. No claim below asserts invariance of Lmin or Lmax under adding singleton values.

Scaling, by contrast, is harmless: $\mathrm{Core}(cv) = c\,\mathrm{Core}(v)$ for $c > 0$ and both selections are positively homogeneous, so $\mathrm{Lmin}(cv) = c\,\mathrm{Lmin}(v)$ and $\mathrm{Lmax}(cv) = c\,\mathrm{Lmax}(v)$. Hence $\rho(v) = \|\,\mathrm{Lmin}(v) - \mathrm{Lmax}(v)\|_1 / v(N)$ is scale-invariant, which is what makes it comparable across games of different size. We record the trivial ceiling and the form of the deficit.

Lemma 3 (ceiling and overlap)**.** *For any singleton-normalized exact v with $T = v(N) > 0$,*

$$\rho(v) = 2\Big(1 - \frac{1}{T}\sum_{i\in N} \min\big(\mathrm{Lmin}(v)_i, \mathrm{Lmax}(v)_i\big)\Big) \;\le\; 2,$$

with $\rho(v) = 2$ if and only if $\mathrm{Lmin}(v)$ and $\mathrm{Lmax}(v)$ have disjoint supports.

Proof. For nonnegative x, y with $x(N) = y(N) = T$ one has $|x_i - y_i| = x_i + y_i - 2\min(x_i, y_i)$, and summing gives $\|x - y\|_1 = 2T - 2\sum_i \min(x_i, y_i)$. The overlap term is nonnegative and vanishes exactly when no coordinate is positive in both. □

So $\rho \to 2$ is precisely asymptotic support disjointness of the two egalitarian selections, and the whole content of the paper is that this can be forced.

3 The two-generator class

Throughout this section v is the lower envelope of exactly two nonnegative allocations g, h with common total T, so that $v(S) = \min(g(S), h(S))$ for every S (including $S = N$, where both sides equal T).

Lemma 4 (support disjointness)**.** *If v is singleton-normalized then $A := \mathrm{supp}(g)$ and $B := \mathrm{supp}(h)$ are disjoint, and every player outside $A \cup B$ is null, receiving 0 in every core allocation. Deleting null players changes neither* Lmin*,* Lmax*, nor T, so we may and do assume $A \cup B = N$ and $A \cap B = \emptyset$.*

Proof. $v(\{i\}) = \min(g_i, h_i) = 0$ forces $g_i = 0$ or $h_i = 0$, which is exactly $A \cap B = \emptyset$. If $i \notin A \cup B$ then $g_i = h_i = 0$, so $v(N \setminus \{i\}) = \min\big(g(N \setminus \{i\}), h(N \setminus \{i\})\big) = T$; any $x \in \mathrm{Core}(v)$ has $x(N \setminus \{i\}) \ge T$ and $x(N) = T$ with $x \ge 0$, hence $x_i = 0$. Both selections assign a null player 0, so deletion is harmless. □

Definition 5. *For $S \subseteq N$ put $f(S) := \max\big(g(S), h(S)\big) = \max\big(g(S \cap A), h(S \cap B)\big)$.*

3.1 A dual description of the core

Theorem 6 (two-generator core duality)**.**

$$\mathrm{Core}(v) = \big\{x \ge 0 :\; x(N) = T \text{ and } x(S) \le f(S) \text{ for all } S \subseteq N\big\}.$$

Proof. For x with $x(N) = T$ and any S, the inequality $x(S) \ge v(S)$ is equivalent to $x(N \setminus S) \le T - v(S)$. It therefore suffices to prove

$$T - v(S) = f(N \setminus S) \qquad \text{for every } S \subseteq N. \tag{1}$$

Since g and h are modular with $g(N) = h(N) = T$,

$$T - v(S) = T - \min\big(g(S), h(S)\big) = \max\big(T - g(S),\, T - h(S)\big) = \max\big(g(N \setminus S),\, h(N \setminus S)\big) = f(N \setminus S).$$

Replacing S by $N \setminus S$ in (1) turns the family $\{x(S) \geq v(S)\}$ into the family $\{x(S) \leq f(S)\}$. Finally $x \geq 0$ is exactly the family of singleton inequalities $x(\{i\}) \geq v(\{i\}) = 0$. □

Identity (1) uses only modularity and the common total; the disjointness of Lemma 4 is not needed for it, and enters only to normalize the presentation of f. Note also that $S = N$ gives $f(N) = T$, so the cap at N is implied by efficiency, and $S = \emptyset$ gives $f(\emptyset) = 0$.

In fact the duality is fully general: for *any* singleton-normalized game, setting $f(S) := T - v(N \setminus S)$, the same complementation involution gives $\mathrm{Core}(v) = \{x \geq 0 : x(N) = T,\ x(S) \leq f(S)\ \forall S\}$. The two-generator content of Theorem 6 is the computation $f = \max(g, h)$. The general form is used in Section 5.4.

3.2 The leximin floor

Theorem 7 (leximin floor: the easy direction)**.** *Let*

$$\lambda := \min_{\emptyset \neq S \subseteq N} \frac{f(S)}{|S|} = \min_{(A', B') \neq (\emptyset, \emptyset)} \frac{\max\big(g(A'), h(B')\big)}{|A'| + |B'|},$$

the second minimum running over $A' \subseteq A$, $B' \subseteq B$. Then $\mu \leq \lambda$.

Proof. $\mathrm{Lmin}(v)$ is a core allocation, so by Theorem 6, for every nonempty S, $|S|\,\mu \leq \mathrm{Lmin}(v)(S) \leq f(S)$, i.e. $\mu \leq f(S)/|S|$. Take the minimum over S. □

The bound is a *one-coalition certificate*: a single tightest cap already pins the floor from above. Its natural habitat is polymatroid theory: when f is monotone submodular, the polytope of Theorem 6 is a polymatroid base polytope, the leximin floor *equals* λ, and the decreasingly-minimal and increasingly-maximal points coincide — the theory of lexicographically optimal bases (Fujishige, 1980), where $\rho = 0$ identically. The two-generator $f = \max(g, h)$ is not submodular in general; the conjecture below asks whether the floor identity survives the loss of submodularity, and the rest of the paper measures exactly how far the dec-min/inc-max coincidence fails. Empirically the bound is exact.

Conjecture 8 (leximin floor formula)**.** *$\mu = \lambda$ for every singleton-normalized two-generator exact game.*

Evidence and reduction. Conjecture 8 holds in 586 of 586 random two-generator games at $n = 4, 5, 6, 7$ (sample sizes 120, 218, 165, 83), with all arithmetic exact. What remains to prove is a clean linear-programming statement. Equality means that the system

$$y \geq 0, \qquad y(N) = T - n\lambda, \qquad y(S) \leq f'(S) := f(S) - \lambda|S|$$

is feasible; indeed, given such y, the allocation $x = y + \lambda\mathbf{1}$ satisfies $x(N) = T$, $x \geq \lambda\mathbf{1} \geq 0$ and $x(S) \leq f(S)$, hence lies in the core with floor at least λ. Note $f' \geq 0$ by definition of λ. By linear-programming duality,

$$\max\{y(N) :\ y \geq 0,\ y(S) \leq f'(S)\ \forall S\} = \min\Big\{\textstyle\sum_S w_S f'(S) :\ w \geq 0,\ \sum_{S \ni i} w_S \geq 1\ \forall i\Big\},$$

the minimum running over fractional covers. Taking $w = \mathbf{1}_{\{S=N\}}$ shows the minimum is at most $f'(N) = T - n\lambda$, so Conjecture 8 is exactly the assertion

$$\sum_S w_S f'(S) \ \geq\ f'(N) \qquad \text{for every fractional cover } w. \tag{2}$$

Proposition 9 (balanced covers)**.** *Inequality (2) holds for every exact (balanced) cover, i.e. whenever* $\sum_{S\ni i} w_S = 1$ *for all* i*.*

Proof. Balancedness gives $\sum_S w_S|S| = \sum_i \sum_{S\ni i} w_S = n$, so, using $f \geq g$ and the modularity of g,

$$\sum_S w_S f'(S) = \sum_S w_S f(S) - \lambda n \;\geq\; \sum_S w_S g(S) - \lambda n = \sum_i g_i \sum_{S\ni i} w_S - \lambda n = g(N) - \lambda n = f'(N),$$

the last step because $g(N) = T$ and $f'(N) = T - \lambda n$. The same computation works with h in place of g. □

What is missing is the case of covers that strictly over-cover some players. Writing $e_i := \sum_{S\ni i} w_S - 1 \geq 0$, the same computation gives

$$\sum_S w_S f'(S) \;\geq\; f'(N) + \max\Big(\sum_{i\in A} g_i e_i,\; \sum_{i\in B} h_i e_i\Big) - \lambda \sum_{i\in N} e_i,$$

and since $\lambda \leq f(\{i\})$, which equals g_i for $i \in A$ and h_i for $i \in B$, each inner sum dominates λ times its *own* block's excess but not the other block's. Closing that gap, or constructing the attaining core allocation directly, is the remaining work.

3.3 The leximax-min ceiling is not dual

The mirror-image statement is an inequality but not an identity, and this is the first place where the two selections part company.

Observation 10 (M-side asymmetry)**.** *For every core allocation, and in particular for* $\mathrm{Lmax}(v)$*,*

$$M \;\geq\; \max_{\substack{P\subseteq A,\; Q\subseteq B \\ (P,Q)\neq(\emptyset,\emptyset)}} \frac{\min\big(g(P), h(Q)\big)}{|P| + |Q|}. \tag{3}$$

Inequality (3) is not an identity: it fails strictly already at $n = 6$*.*

Proof of the inequality. Let $x \in \mathrm{Core}(v)$ and put $S = P \cup Q$. Then $g(S) = g(P)$ and $h(S) = h(Q)$ because g vanishes on B and h on A, so

$$\max_i x_i \;\geq\; \frac{x(S)}{|S|} \;\geq\; \frac{v(S)}{|S|} = \frac{\min\big(g(P), h(Q)\big)}{|P| + |Q|}.$$

Apply this to $x = \mathrm{Lmax}(v)$. Equivalently, (3) is Theorem 6 read on complements. □

How to read (3). The maximum runs over pairs (P, Q) other than $(\emptyset, \emptyset)$, with the convention $g(\emptyset) = h(\emptyset) = 0$. There is no infinity convention: a pair with an empty part contributes $\min(\cdot, 0) = 0$ and is simply inactive, so only pairs with *both* parts nonempty can bind. This is visible in the proof, where an empty part makes $v(S)$ the minimum of something and 0. Reading an empty part as $+\infty$ instead would reduce (3) to a maximum over singletons and give a different — and wrong — value; the misreading is easy enough that we spell the convention out.

The smallest counterexample. Scanning the complete two-generator enumeration in increasing T, no counterexample to equality in (3) exists at $n \le 5$ for any T, nor at $n = 6$ for $T \le 4$. At $n = 6$, $T = 5$:

$$g = (3, 2, 0, 0, 0, 0), \qquad h = (0, 0, 2, 1, 1, 1),$$
$$\mathrm{Lmin}(v) = \left(1, \tfrac{2}{3}, \tfrac{4}{3}, \tfrac{2}{3}, \tfrac{2}{3}, \tfrac{2}{3}\right), \qquad \mathrm{Lmax}(v) = \left(\tfrac{6}{5}, \tfrac{4}{5}, \tfrac{6}{5}, \tfrac{3}{5}, \tfrac{3}{5}, \tfrac{3}{5}\right),$$

so $M = 6/5$, whereas the right-hand side of (3) equals 1, attained for instance at $P = \{1\}, Q = \{3\}$ with $\min(3, 2)/2 = 1$. The gap is $1/5$. One further counterexample class occurs at $n = 6$, $T = 5$: $g = (3, 1, 1)$, $h = (2, 2, 1)$ on the respective supports (its block-swapped relabelling is the same game up to permutation), again with $M = 6/5$ against a bound of 1. Over the same random samples used for Conjecture 8, (3) is tight in 120/120 games at $n = 4$, 218/218 at $n = 5$, 160/165 at $n = 6$ and 74/83 at $n = 7$.

Remark 11 (diagnosis)**.** *Theorem 7 bounds the floor using only the upper caps $x(S) \le f(S)$, and a single tightest cap can be saturated on its own, which is why one coalition certifies the optimum. Bounding the ceiling instead requires the lower bounds $x(N \setminus S) \ge T - f(S)$, and* minimizing *a maximum must respect several of them simultaneously; no single one certifies the optimum. This is a genuine asymmetry between the two selections, and it is the reason* Lmax *is the harder object throughout.*

4 The master family

Definition 12. *For integers $p \ge 1$ and $k \ge 2$ put $n = p + 1 + k$ and $T = pk$, and let $G(p, k)$ be the lower envelope of*

$$g = \big(p(k-1),\ \underbrace{1, \dots, 1}_{p},\ \underbrace{0, \dots, 0}_{k}\big), \qquad h = \big(\underbrace{0, \dots, 0}_{p+1},\ \underbrace{p, \dots, p}_{k}\big).$$

We index the players $0, 1, \dots, p + k$ and call player 0 the head*, $O := \{1, \dots, p\}$ the* ones*, and $B := \{p + 1, \dots, p + k\}$; so $A = \{0\} \cup O$ has $p + 1$ elements and $|B| = k$.*

Both generators are nonnegative with total $p(k-1)+p = pk = T$, their supports are disjoint, and every coordinate vanishes in one of them; by Lemma 1, $G(p, k)$ is exact and singleton-normalized.

Theorem 13 (closed forms)**.** *For all $p \ge 1$, $k \ge 2$,*

$$\mathrm{Lmin}\big(G(p,k)\big) = \left(\tfrac{p^2(k-1)}{p+1},\ \left[\tfrac{p}{p+1}\right]^{p+k}\right),$$
$$\mathrm{Lmax}\big(G(p,k)\big) = \left(\tfrac{p(k-1)}{k},\ \left[\tfrac{1}{k}\right]^{p},\ \left[\tfrac{p(k-1)}{k}\right]^{k}\right),$$
$$\big\|\mathrm{Lmin} - \mathrm{Lmax}\big\|_1 = \frac{2p\big(p(k-1)-1\big)}{p+1}, \qquad \rho\big(G(p,k)\big) = \frac{2\big(p(k-1)-1\big)}{k(p+1)},$$

where the exponents indicate the number of coordinates, listed in the order head, ones, B.

The proof occupies Sections 4.1–4.6: a key cap, the two optimizations, a membership check, and the arithmetic of the gap.

4.1 The key cap

For each $b \in B$,

$$f\big(O \cup \{b\}\big) = \max\big(g(O), h(\{b\})\big) = \max(p, p) = p,$$

so by Theorem 6 every core allocation x satisfies

$$x_1 + \cdots + x_p + x_b \;\le\; p \qquad \text{for every } b \in B. \tag{KC}$$

That single family of $(p+1)$-player caps drives everything below. Note that the two generators *tie* at these coalitions: the maximum defining f is attained on both sides.

4.2 Leximin

Let $x \in \mathrm{Core}(G(p,k))$ and $m := \min_i x_i$. Applying (KC) with any fixed b gives $(p+1)m \le p$, so

$$m \le \frac{p}{p+1}, \qquad \text{hence} \qquad \mu \le \frac{p}{p+1}.$$

Suppose $m = p/(p+1)$. Then in (KC) the $p+1$ numbers $x_1, \ldots, x_p, x_b$ are each at least $p/(p+1)$ and sum to at most $p = (p+1)\cdot p/(p+1)$, so all of them equal $p/(p+1)$. This holds for *every* $b \in B$, hence

$$x_1 = \cdots = x_p = x_{p+1} = \cdots = x_{p+k} = \frac{p}{p+1},$$

and efficiency forces the head to

$$x_0 = T - (p+k)\frac{p}{p+1} = pk - \frac{p(p+k)}{p+1} = \frac{p^2(k-1)}{p+1},$$

using $k - (p+k)/(p+1) = (kp+k-p-k)/(p+1) = p(k-1)/(p+1)$. Thus the max-min allocation is *unique*, and a unique max-min allocation is a fortiori the leximin allocation. Since $p(k-1) \ge 1$, the head's coordinate is at least $p/(p+1)$, so the floor of this allocation is indeed $p/(p+1)$. It remains to check that this point is in the core, which is done in Section 4.4.

4.3 Leximax-min

Let $x \in \mathrm{Core}(G(p,k))$, $M := \max_i x_i$ and $s := x_1 + \cdots + x_p$. By (KC), $x_b \le p - s$ for every $b \in B$, and trivially $x_b \le M$ and $x_0 \le M$. Efficiency gives

$$pk = x_0 + s + \sum_{b \in B} x_b \;\le\; M + s + k\min(M,\, p - s). \tag{4}$$

Case A: $M \le p - s$. Then (4) reads $pk \le (k+1)M + s$. The case hypothesis is $s \le p - M$, so $pk \le (k+1)M + p - M = kM + p$, i.e. $M \ge p(k-1)/k$.

Case B: $M > p - s$. Then (4) reads $pk \le M + s + k(p-s)$, i.e. $(k-1)s \le M$, so $s \le M/(k-1)$. The case hypothesis then gives $M > p - s \ge p - M/(k-1)$, i.e. $Mk/(k-1) > p$, i.e. $M > p(k-1)/k$.

In both cases $M \ge p(k-1)/k$, so the leximax-min ceiling is at least $p(k-1)/k$. Equality can occur only in Case A with every inequality used tight, that is with $x_0 = M$, with $x_b = M$ for all $b \in B$, and with

$$s = pk - (k+1)M = pk - (k+1)\frac{p(k-1)}{k} = \frac{p\big(k^2 - (k+1)(k-1)\big)}{k} = \frac{p}{k}.$$

(Consistency: $p - s = p - p/k = p(k-1)/k = M$, so Case A holds with equality, as it must.) Thus at the min-max level the head and all of B are pinned at $p(k-1)/k$ and the ones share exactly p/k; since $p/k \le p(k-1)/k$ for $k \ge 2$, the ones occupy the last p positions of the decreasing rearrangement. The next leximax stage minimizes their maximum subject to their sum being p/k, whose unique optimum is $1/k$ each. Hence Lmax is as claimed, again modulo membership.

4.4 Membership

This step is load-bearing: the two optimizations above determine the optima only conditionally on the described allocations being feasible.

Index as in Definition 12 and, for $S \subseteq N$, put

$$\varepsilon := [\,0 \in S\,], \qquad i := |S \cap O|, \qquad j := |S \cap B|,$$

so that $0 \le i \le p$, $0 \le j \le k$ and

$$g(S) = \varepsilon\, p(k-1) + i, \qquad h(S) = jp, \qquad f(S) = \max\big(\varepsilon\, p(k-1) + i,\ jp\big). \tag{5}$$

Lemma 14. *Both allocations of Theorem 13 lie in* $\mathrm{Core}(G(p,k))$ *for all* $p \ge 1$, $k \ge 2$. *Moreover, for both of them the tight coalitions are exactly*

$$\begin{array}{lll} \textit{(T1)} & S = O \cup \{b\},\ b \in B, & f(S) = \max(p,p) = p, \\ \textit{(T2)} & S = \{0\} \cup B',\ |B'| = k-1, & f(S) = \max\big(p(k-1),(k-1)p\big) = p(k-1), \\ \textit{(T3)} & S = N, & f(N) = T. \end{array}$$

Proof. The leximin allocation. Write $q := p/(p+1)$ and $H := p^2(k-1)/(p+1) = p(k-1)q$, so $x^* = (H, q, \dots, q)$ with $p+k$ copies of q. Efficiency:

$$H + (p+k)q = q\big(p(k-1) + p + k\big) = q(pk+k) = k(p+1)\frac{p}{p+1} = pk = T,$$

and $x^* > 0$. It remains to check $x^*(S) = \varepsilon H + (i+j)q \le f(S)$.

$\varepsilon = 0$, $j = 0$: S is a subset of O, and $x^*(S) = iq < i = f(S)$ for $i \ge 1$ because $q < 1$; for $i = 0$, $S = \emptyset$. Strict.

$\varepsilon = 0$, $j \ge 1$: bound $f(S) \ge jp$ and use $i \le p$:

$$x^*(S) = (i+j)\frac{p}{p+1} \le (p+j)\frac{p}{p+1} \le jp \iff p + j \le j(p+1) \iff p \le jp,$$

which holds since $j \ge 1$. Equality throughout requires $i = p$ and $j = 1$, i.e. $S = O \cup \{b\}$, which is (T1).

$\varepsilon = 1$, $j \le k-1$: put $G_0 := p(k-1) + i = g(S)$, so $x^*(S) = q(G_0 + j)$ and

$$x^*(S) \le G_0 \iff p(G_0 + j) \le (p+1)G_0 \iff jp \le G_0 = p(k-1) + i,$$

which holds because $jp \le (k-1)p$. Hence $x^*(S) \le g(S) \le f(S)$, with equality exactly when $j = k-1$ and $i = 0$, i.e. $S = \{0\} \cup B'$ with $|B'| = k-1$, which is (T2).

$\varepsilon = 1$, $j = k$: then $h(S) = kp = T$, so $f(S) \ge T \ge x^*(S)$, with equality only for $S = N$ since $x^* > 0$; this is (T3).

The leximax-min allocation. Write $L := p(k-1)/k$ and $r := 1/k$, so $y^* = (L, r, \dots, r, L, \dots, L)$ with p copies of r and k copies of L. Efficiency:

$$L + pr + kL = (k+1)L + \frac{p}{k} = \frac{p\big[(k+1)(k-1)+1\big]}{k} = \frac{pk^2}{k} = pk = T,$$

and $y^* > 0$. Now $y^*(S) = \varepsilon L + ir + jL$.

$\varepsilon = 0$, $j = 0$: $y^*(S) = i/k \le i = f(S)$ since $k \ge 2$, strictly for $i \ge 1$.

$\varepsilon = 0$, $j \ge 1$: comparing with the h-side,

$$\frac{i + jp(k-1)}{k} \le jp \iff i + jp(k-1) \le jpk \iff i \le jp,$$

which holds since $i \le p \le jp$, with equality exactly when $i = p$ and $j = 1$, i.e. (T1).

$\varepsilon = 1$, $j \le k-1$: $y^*(S) = (j+1)L + i/k$, and multiplying by $k > 0$,

$$y^*(S) \le p(k-1) + i \iff (j+1)p(k-1) + i \le kp(k-1) + ki \iff p(j+1-k) \le i,$$

after dividing by $k - 1 \ge 1$; the left-hand side is at most $0 \le i$, with equality exactly when $j = k-1$ and $i = 0$, i.e. (T2).

$\varepsilon = 1$, $j = k$: as before $f(S) \ge h(S) = kp = T \ge y^*(S)$, with equality only at $S = N$. □

4.5 The hinge: both selections saturate the same caps

Lemma 14 contains the expository point of the paper. The active constraints are the same for Lmin and Lmax: the two families (T1) and (T2), plus efficiency. Both are coalitions at which the two generators *tie*, which is the structural reason they are the active ones — there is no slack on either side to give. What differs is only how the two selections distribute mass *inside* a tight coalition. On $S = O \cup \{b\}$, which has total exactly p:

	each of the p ones	the single $b \in B$
Lmin :	$\frac{p}{p+1}$	$\frac{p}{p+1}$
Lmax :	$\frac{1}{k}$	$\frac{p(k-1)}{k}$

Leximin splits the budget p evenly over all $p+1$ members; leximax-min gives the ones the least it can and puts the whole remainder on the one member of B, because loading B is how it lowers the global maximum. The divergence is visible in this one coalition. Everything else is bookkeeping: leximin's even split leaves the residual $T - (p+k)p/(p+1)$ with nowhere to go but the head, while leximax-min's loading of every $b \in B$ up to $p(k-1)/k$ consumes almost all of T across B.

4.6 The gap

Proof of Theorem 13, conclusion. Put $D := p(k-1) - 1 \ge 0$. Coordinatewise,

$$\text{head:} \quad \frac{p^2(k-1)}{p+1} - \frac{p(k-1)}{k} = \frac{p(k-1)D}{k(p+1)},$$

$$\text{each of the } p \text{ ones:} \quad \frac{p}{p+1} - \frac{1}{k} = \frac{D}{k(p+1)},$$

$$\text{each of the } k \text{ members of } B\text{:} \quad \frac{p(k-1)}{k} - \frac{p}{p+1} = \frac{pD}{k(p+1)},$$

all three nonnegative. Hence

$$\| \text{Lmin} - \text{Lmax} \|_1 = \frac{p(k-1)D + pD + kpD}{k(p+1)} = \frac{D\big(p(k-1) + p + kp\big)}{k(p+1)} = \frac{2pkD}{k(p+1)} = \frac{2pD}{p+1},$$

and $\rho = 2pD/\big((p+1)pk\big) = 2D/\big(k(p+1)\big)$. □

Corollary 15 (deficit identity). $2-\rho\big(G(p,k)\big)=\dfrac{2n}{k(p+1)}$, *where* $n=p+1+k$.

Proof. $2-\dfrac{2\big(p(k-1)-1\big)}{k(p+1)}=\dfrac{2\big[k(p+1)-p(k-1)+1\big]}{k(p+1)}=\dfrac{2(kp+k-pk+p+1)}{k(p+1)}=\dfrac{2(k+p+1)}{k(p+1)}$. □

Remark 16 (the deficit is twice the overlap). *Corollary 15 also follows from Lemma 3. In* $G(p,k)$ *the coordinatewise minima of the two allocations are* $p(k-1)/k$ *at the head,* $1/k$ *at each one, and* $p/(p+1)$ *at each member of* B*, so*

$$\sum_i \min\big(\mathrm{Lmin}_i,\mathrm{Lmax}_i\big)=\frac{p(k-1)}{k}+\frac{p}{k}+\frac{kp}{p+1}=p+\frac{kp}{p+1}<p+k=n-1,$$

and $2\cdot\big(p+kp/(p+1)\big)/(pk)=2(p+1+k)/\big(k(p+1)\big)$. *The overlap between the two selections is thus bounded by* n *while* $T=pk$ *grows quadratically: this is the precise sense in which their supports become asymptotically disjoint.*

4.7 The supremum

Theorem 17 (value of the supremum). *For every* $n\ge 4$,

$$s(n)\;\ge\;2-\frac{8n}{n^2-(n \bmod 2)}=\begin{cases}2-\dfrac{8}{n}, & n \text{ even},\\[2mm] 2-\dfrac{8n}{n^2-1}, & n \text{ odd},\end{cases}$$

attained on the two-generator class by $G(p,k)$ *with the balanced split* $k=\lceil n/2\rceil$, $p+1=\lfloor n/2\rfloor$. *Consequently* $\sup_n s(n)=2$.

Proof. By Corollary 15 the deficit of $G(p,k)$ is $2n/(k(p+1))$ with $k+(p+1)=n$, $k\ge 2$ and $p+1\ge 2$. Minimizing the deficit means maximizing the product $k(p+1)$ of two positive integers with fixed sum n, whose maximum is $\lfloor n/2\rfloor\lceil n/2\rceil$, equal to $n^2/4$ for even n and $(n^2-1)/4$ for odd n, attained at the balanced split (for $n\ge 4$ this split satisfies the constraints $k\ge 2$, $p\ge 1$). Substituting gives the displayed bound. For the last claim, $\rho\le 2$ always by Lemma 3, and $2-8/n\to 2$. □

Table 1 lists the resulting values. The ratio reaches 1 at $n=8$ and passes it at $n=9$: from eight players on the two egalitarian core selections can differ by the entire worth of the grand coalition, and from nine players by more.

Remark 18 (the degenerate corner). $G(1,2)$ *has* $n=4$, $T=2$, $g=(1,1,0,0)$, $h=(0,0,1,1)$ *and* $\rho=2(1\cdot 1-1)/(2\cdot 2)=0$, *consistent with the four-player rigidity; the deficit identity reads* $2-0=2n/(k(p+1))=8/4=2$.

Remark 19 (there is nothing special about $|B|=4$). *An earlier version of this family fixed* $k=|B|=4$, *giving* $\rho=(3p-1)/\big(2(p+1)\big)\to 3/2$. *Corollary 15 shows that* k *is a free parameter and that the optimum is* $k\approx n/2$, *so the limit* $3/2$ *was an artifact of the fixed block size; the* $k=4$ *slice is already beaten from* $n=10$ *on (*$7/6$ *against* $6/5$*). The number* 4 *that appears as* $|\,\mathrm{supp}(\mathrm{Lmax})|$ *in the* $n=5$ *extremal class (Section 5) is a small-*n *phenomenon and not a feature of the asymptotics: in* $G(p,k)$ *every coordinate of* Lmax *is strictly positive, so* $|\,\mathrm{supp}(\mathrm{Lmax})|=n$ *and no player is starved at all.*

n	(p,k)	lower bound on $s(n)$	decimal
6	$(2,3)$	$2/3$	0.6667
7	$(2,4)$	$5/6$	0.8333
8	$(3,4)$	1	1.0000
9	$(3,5)$	$11/10$	1.1000
10	$(4,5)$	$6/5$	1.2000
11	$(4,6)$	$19/15$	1.2667
12	$(5,6)$	$4/3$	1.3333
16	$(7,8)$	$3/2$	1.5000
20	$(9,10)$	$8/5$	1.6000
100	$(49,50)$	$48/25$	1.9200

Table 1: The bound of Theorem 17 at the balanced split. All entries are exact; decimals are rounded.

n	totals	games	enumerated max ρ	Theorem 17
3	2–30	225	0	—
4	2–30	3,040	0	0
5	2–24	4,357	$1/2$	$1/3$
6	2–20	8,213	$2/3$	$2/3$
7	2–14	1,609	$5/6$	$5/6$
8	2–12	1,049	1	1

Table 2: Complete two-generator enumerations. The rows $n = 3, 4$ are controls: they reproduce the four-player rigidity. The last column is the value of the balanced-split family bound.

5 Fixed n: enumeration, certificates, and the $n = 5$ anomaly

5.1 Complete two-generator enumeration

Write $s_2(n)$ for the supremum of ρ over singleton-normalized two-generator exact games on n players. Table 2 reports complete enumerations, up to player permutation, generator swap and scaling, of all integer generator pairs with common total in the stated range. All arithmetic is exact rational.

Two readings must be kept apart.

What is complete. Because ρ is scale-invariant, every two-generator game with rational generators has a unique primitive integer representative, and enumeration to total $T_{\max}$ is exactly complete for the games whose primitive representative has total at most $T_{\max}$ — equivalently, for rational shapes of denominator at most $T_{\max}$. It says nothing about finer denominators or about irrational generators, so enumeration-to-T is a lower-bound method for $s_2(n)$: the entries of Table 2 are maxima over the enumerated range, not values of s_2. As a partial probe of the excluded region, shape hill-climbing at fine resolution was run from the known witnesses (resolution $1/60$ at $n = 5$; $1/56$ and $1/60$ at $n = 6$; $1/64$ at $n = 7$). None improved on the tabulated maximum; at $n = 7$ the climb from the $5/6$ witness stalled after 42 steps, and a second start from a base of $2/3$ climbed only to $3/4$.

What agrees. The enumerated maxima agree exactly with Theorem 17 at $n = 6, 7, 8$, so on the two-generator class that bound is attained there, and

$$s_2(n) \stackrel{?}{=} 2 - \frac{8n}{n^2 - (n \bmod 2)} \qquad (n \geq 6)$$

is a live candidate, currently confirmed at three values of n. At $n = 5$ the two disagree, and the enumeration is the correct one; see Section 5.3.

5.2 Searching the full exact class

The two-generator restriction is not innocuous. Adding a generator *lowers* the envelope and hence *grows* the core, so it is not automatic that two generators are extremal. The following observation lets one search the entire exact class rather than a bounded-generator slice.

Lemma 20 (core closing)**.** *Fix $T > 0$ and a vector $c = (c_S)$ indexed by the proper coalitions, and put $P(c) = \{x : x(N) = T,\ x(S) \geq c_S$ for all proper $S\}$. If $P(c) \neq \emptyset$, then the game v defined by $v(S) = \min_{x \in P(c)} x(S)$ for proper S and $v(N) = T$ is exact, and $\mathrm{Core}(v) = P(c)$.*

Proof. Every $x \in P(c)$ satisfies $x(S) \geq v(S)$ by definition of v and $x(N) = T$, so $P(c) \subseteq \mathrm{Core}(v)$; and each minimum is attained, so v is exact. Conversely, $v(S) \geq c_S$ for every proper S because every point of $P(c)$ satisfies $x(S) \geq c_S$; hence $x \in \mathrm{Core}(v)$ implies $x(S) \geq v(S) \geq c_S$ and $x(N) = T$, i.e. $x \in P(c)$. □

Perturbing c and re-closing therefore moves within the exact class and reaches cores with arbitrarily many vertices. Hill climbing in this "c-space", *seeded at the extremal two-generator game* of Table 2, produced no improvement at $n = 5$ (240 steps, stays at $1/2$), $n = 6$ (496 steps, stays at $2/3$) or $n = 7$ (301 steps, stays at $5/6$). Since the search is over all exact games, this certifies the two-generator optimum as a *local* optimum in the full exact class at those three values of n. Independently, from the known extremal at each of $n = 5, 6, 7, 8$, about 200 structured and random third generators at fixed common total were added: zero improvements.

One limitation must be stated. Unseeded c-space restarts reach only $1/3$ at $n = 6$, far below the known $2/3$; that variant of the search is under-powered for $n \geq 6$ and certifies nothing. Only the seeded runs support the local-optimality claim, and none of these runs bears on global optimality.

5.3 The $n = 5$ anomaly

At $n = 5$ the balanced-split family gives only $1/3$, but the truth is $1/2$, and the witness has a non-uniform h-block. The proof is by hand from two flat splits of the game; no linear program and no vertex enumeration is used.

Theorem 21 ($n = 5$)**.** *Let w be the lower envelope of $g^1 = (1, 0, 0, 1, 2)$ and $g^2 = (0, 1, 3, 0, 0)$. Then w is exact and singleton-normalized with $w(N) = 4$, and*

$$\mathrm{Lmin}(w) = \left(\tfrac{1}{2}, \tfrac{1}{2}, \tfrac{5}{4}, \tfrac{1}{2}, \tfrac{5}{4}\right), \qquad \mathrm{Lmax}(w) = (1, 0, 1, 1, 1), \qquad \| \mathrm{Lmin} - \mathrm{Lmax} \|_1 = 2, \qquad \rho(w) = \tfrac{1}{2}.$$

In particular $s(5) \geq 1/2 > 1/3$.

Proof. Exactness and normalization are immediate from Lemma 1, each coordinate vanishing in one generator. Two flat splits control the core: $w(\{1, 2\}) = 1$ and $w(\{3, 4, 5\}) = 3$ sum to $w(N) = 4$, so

$x_1 + x_2 = 1$ on all of Core(w), and likewise $w(\{2,4\}) = 1$ with $w(\{1,3,5\}) = 3$ gives $x_2 + x_4 = 1$. Putting $s := x_1$, every core point therefore satisfies

$$x_1 = x_4 = s, \qquad x_2 = 1 - s, \qquad x_3 + x_5 = 3 - s, \tag{6}$$

with $s \in [0,1]$ by nonnegativity of x_1 and x_2.

Leximin. By (6), $\min_i x_i \le \min(s, 1-s) \le 1/2$, with equality forcing $s = 1/2$ and hence $x_1 = x_2 = x_4 = 1/2$. Then $x_3 + x_5 = 5/2$, and the leximin continuation splits this evenly, giving $5/4$ each. The resulting allocation $x = (1/2, 1/2, 5/4, 1/2, 5/4)$ lies in the core, as direct substitution against both generators confirms. Hence Lmin(w) is as claimed.

Leximax-min. By (6), $\max_i x_i \ge (x_3 + x_5)/2 = (3-s)/2 \ge 1$ since $s \le 1$. Equality forces $s = 1$, so $x = (1, 0, x_3, 1, x_5)$ with $x_3 + x_5 = 2$; the ceiling $\max_i x_i = 1$ then forces $x_3 = x_5 = 1$. The unique candidate $(1,0,1,1,1)$ lies in the core, so Lmax(w) $= (1,0,1,1,1)$.

Finally $\| \text{Lmin} - \text{Lmax} \|_1 = \frac{1}{2} + \frac{1}{2} + \frac{1}{4} + \frac{1}{2} + \frac{1}{4} = 2$ and $\rho = 2/4 = 1/2$. □

Four independent methods agree on this game: the hand argument above, exact rational core-vertex enumeration, a staged exact solver over linear real arithmetic, and a from-scratch exact-rational two-phase simplex with Bland's rule.

The value $\frac{1}{2}$ is in fact the supremum: $s(5) = \frac{1}{2}$ (Theorem 33 below, by a configuration-complete exact finite check). Before that machinery, the searches already pointed there: roughly 4×10^4 distinct exact five-player games evaluated in exact rational arithmetic produced nothing above $1/2$ — the complete two-generator enumeration of Table 2, complete three-generator enumerations at totals 4–7, c-space hill climbing in the sense of Lemma 20, and fine-denominator shape climbing from the extremal witness — with the $n = 3, 4$ controls returning maximum 0 from the same pipeline.

Structure of the extremal class. The following are computational findings over 360 distinct profiled games at $n = 5$, of which 199 have $\rho > 1/3$ and 107 attain $\rho = 1/2$. Among the 107 extremal games, without exception:

- Lmin has stratum composition $(3,2)$, and Lmax has stratum composition $(1,4)$;
- Lmax has exactly one zero coordinate, so $|\text{supp}(\text{Lmax})| = 4$;
- normalized to $T = 4$ the pair is unique up to permutation: Lmin $\sim (\frac{1}{2}, \frac{1}{2}, \frac{1}{2}, \frac{5}{4}, \frac{5}{4})$ and Lmax $\sim (0,1,1,1,1)$ — that is, exactly the pair of Theorem 21.

Among the 199 games above $1/3$, the Lmin strata are $(3,2)$ in 196 cases and $(2,1,2)$ in 3; the pattern $(4,1)$ never occurs above $1/3$, and Lmax is never the uniform allocation.

One tempting invariant is refuted. Call S a *flat split* if $v(S) + v(N \setminus S) = v(N)$; then $x(S)$ is constant on the core, which is the affine seesaw visible in (6) — in the game of Theorem 21, $w(\{1,2\}) + w(\{3,4,5\}) = 1 + 3 = 4$. Only 76 of the 107 extremal games have any flat split, and 31 attain exactly $1/2$ with none, so the seesaw is a feature of the particular presentation in Theorem 21 and not an invariant of extremality. When a flat split does exist its sizes are always $\{2,3\}$, and the Lmax-starved player lies on the size-2 side in all 76 cases.

Two regimes. Theorem 17 gives only $1/3$ at $n = 5$ because the family $G(p,k)$ has a uniform h-block and the available splits $(k, p+1) \in \{(2,3),(3,2)\}$ are too coarse; the extremal five-player game has a non-uniform h-block. The crossover to the G-family shape happens at $n = 6$. The two regimes differ qualitatively and not just numerically: at $n = 5$ the extremal Lmax starves one player to 0 and has support of size 4, whereas in $G(p,k)$ the extremal Lmax is strictly positive everywhere. Explaining the crossover is open (Section 7).

5.4 One-coalition certificates and the regime change

Write, for any singleton-normalized exact game with the general dual bound $f(S) = T - v(N \setminus S)$,

$$\lambda := \min_{\emptyset \neq S} \frac{f(S)}{|S|}, \qquad \nu := \max_{\emptyset \neq S} \frac{v(S)}{|S|}, \qquad \mu := \min_i \mathrm{Lmin}(v)_i, \qquad M := \max_i \mathrm{Lmax}(v)_i.$$

The general duality gives the bracketing $\mu \leq \lambda$ and $M \geq \nu$ at once (for $\mu \leq \lambda$ the proof of Theorem 7 applies verbatim). On a two-generator game the right-hand side of (3) is exactly ν, since $v(P \cup Q) = \min(g(P), h(Q))$. Observation 10 says the second bracket can be strict from $n = 6$ on. Two facts sharpen this into a statement about the regime change at five players.

Proposition 22 (covering obstruction)**.** *Let $\tau_\nu := \{S : v(S) = \nu|S|\}$ be the ν-tight family. If τ_ν covers N and $\nu > T/n$, then no core point satisfies $x \leq \nu\mathbf{1}$, and hence $M > \nu$.*

Proof. Sum the core inequalities $x(S) \geq v(S) = \nu|S|$ over a cover with weights $w \geq 0$: with $c_i := \sum_{S \ni i} w_S \geq 1$ this gives $\sum_i c_i(x_i - \nu) \geq 0$. If $x \leq \nu\mathbf{1}$, every term is nonpositive, so $x_i = \nu$ for every i; then $T = \nu n$, contradicting $\nu > T/n$. □

Moreover no *balanced* collection can witness $M > \nu$: for balanced w, $\sum_S w_S v(S) \leq T$ because the core is nonempty, and the same computation then yields $\sum_S w_S\big(\nu|S| - v(S)\big) \geq \nu n - T$, which is what equality requires. So a failure of $M = \nu$ needs a strictly over-covering fractional cover: no balanced collection can witness one. A covering ν-tight family is one way this arises, and it is what occurs at the smallest counterexample of Observation 10: the four ν-tight sets $\{2,3\}, \{1,3,4\}, \{1,3,5\}, \{1,3,6\}$ cover N with per-player excess $(2,0,3,0,0,0)$, and player 2, lying only in $\{2,3\}$, blocks every balanced subfamily. At the five-player extremal game of Theorem 21, by contrast, $\nu = 1$ and the ν-tight family $\{3,5\}, \{1,3,5\}, \{3,4,5\}$ misses player 2, so Proposition 22 does not apply, and the dual minimum $h(N) = 1$ is attained at the balanced partition $\{2,4\} \mid \{1,3,5\}$. The covering test is sufficient for $M > \nu$ but not necessary: at six players, games with $M > \nu$ outnumber those with a covering ν-tight family.

Theorem 23 (one-coalition certificates at five players)**.** *For every singleton-normalized exact game on five players, $\mu = \lambda$ and $M = \nu$: both egalitarian levels are certified by a single coalition.*

Proof, by finite exact check. By the duality above, $M = \nu$ fails if and only if some fractional cover w has $\sum_S w_S h(S) < h(N)$, where $h(S) = \nu|S| - v(S) \geq 0$; likewise for μ with $f'(S) = f(S) - \lambda|S|$. Both objectives are nonnegative on the cover polyhedron, which is pointed with recession cone the nonnegative orthant, so each minimum is attained at a vertex; on five players there are 2287 vertices, each of support at most five. Fixing a vertex and a coalition attaining ν (resp. λ) makes every hypothesis, and the failure inequality, linear in v, and $v(N) = 1$ is free by homogeneity. Exactness is imposed through the 280 inequalities that the facet theorem of Studený and Kratochvíl (2022) assigns to the sixteen indecomposable min-semi-balanced systems on five players under all permutations; this description was validated in both directions against independent exactness tests. Each of the 1142 symmetry classes of (vertex, anchor) pairs, on each side, is infeasible in exact rational arithmetic. □

Two remarks calibrate the proof. First, deleting the 280 exact-cone inequalities makes 361 (resp. 339) of the over-covering classes feasible, so it is exactness — not the shape of the certificate — that forces the one-coalition property at five players. Second, the M-side is false from six players on (Observation 10), and the same reduction detects that failure. The theorem is therefore the formal content of the regime change of Section 5.3: at five players both selections are governed by

single bottleneck coalitions, while from six players the ceiling requires genuinely fractional covers — which is precisely where $s(n)$ begins its climb toward 2. One consequence localizes all the difficulty of $s(5)$.

Corollary 24 (floor from ceiling)**.** *For every singleton-normalized exact five-player game,* $\mu \geq \min_i \mathrm{Lmax}(v)_i$.

Proof. Lmax is a core point, so $f(S) = T - v(N \setminus S) \geq T - \mathrm{Lmax}(N \setminus S) = \mathrm{Lmax}(S)$ for every S; hence $\lambda \geq \min_S \mathrm{Lmax}(S)/|S| \geq \min_i \mathrm{Lmax}_i$, and $\mu = \lambda$ by Theorem 23. □

If Lmax has no zero coordinate, the leximin floor is therefore bounded below for free and elementary gap arithmetic bounds the ratio; every genuinely hard configuration for $s(5)$ has a *zero* in Lmax. The theorem does *not* by itself deliver the starvation bound needed for those configurations; that bound is a statement about a different functional and requires its own argument, which the next theorem supplies in the extremal one.

Theorem 25 (the extremal configuration is pinned)**.** *Let v be a singleton-normalized exact five-player game whose* Lmin *strata form the pattern* $(3, 2)$ *and whose* Lmax *strata form the pattern* $(1, 4)$ *with the singleton stratum at zero, at player j. Then*

$$f(\{j\}) = T - v(N \setminus \{j\}) \;\geq\; \tfrac{T}{8}.$$

Proof. Normalize $T = 1$ and suppose $f(\{j\}) < \frac{1}{8}$. The starved player j is Lmin-poor by Corollary 30 below (whose two-line proof uses only Lemma 29 and is independent of this theorem); write $P = \{j, p_1, p_2\}$ for the poor stratum and $R = \{r_1, r_2\}$ for the rich pair, with $x =$ Lmin equal to μ on P and $\varrho = (1 - 3\mu)/2$ on R, and $z =$ Lmax equal to 0 at j and $\frac{1}{4}$ elsewhere. Call S *f-tight* when $x(S) = f(S)$, equivalently when $N \setminus S$ is v-tight at x.

Step 0. $\lambda \leq f(\{j\}) < \frac{1}{8}$, and $\mu \leq \lambda$ holds for every game (the easy half of Theorem 23, i.e. Theorem 7's argument), so $\mu < \frac{1}{8}$; hence $\varrho > \frac{5}{16} > \frac{1}{4}$ and $c := 1 - 5\mu > 0$.

Step 1 (key lemma). Every f-tight S satisfies $|S \cap \{p_1, p_2\}| \leq |S \cap R|$. Indeed, $U = N \setminus S$ is v-tight, and $z \in \mathrm{Core}(v)$ gives, with $a = |U \cap P|$, $b = |U \cap R|$, $\varepsilon = [\, j \in U \,]$,

$$\mu a + \varrho b \;=\; x(U) = v(U) \;\leq\; z(U) \;=\; \tfrac{a+b-\varepsilon}{4}.$$

The claim is $b \leq a - \varepsilon$. A violation leaves four shapes, each contradictory: $a - \varepsilon = 0$ with $\varepsilon = 0$ gives $\varrho \leq \frac{1}{4}$; with $\varepsilon = 1$ it gives $\mu \leq b(\frac{1}{4} - \varrho) < 0$; $a - \varepsilon = 1$ with $\varepsilon = 1$ (so $b = 2$) gives $1 - \mu \leq \frac{3}{4}$, i.e. $\mu \geq \frac{1}{4}$; and $a - \varepsilon = 1$ with $\varepsilon = 0$, $b = 2$ — the coalition $U = \{p_i\} \cup R$ — gives $1 - 2\mu \leq \frac{3}{4}$, i.e. $\mu \geq \frac{1}{8}$, against the failure hypothesis. This last case is the only one that mentions $\frac{1}{8}$.

Step 2 (the pinning, dualized). That P is the Lmin bottom stratum means every core point $y \geq \mu\mathbf{1}$ has $y(P) = 3\mu$. Substituting $u = y - \mu\mathbf{1}$, this is $\max\{u(P) : u \geq 0,\ u(N) = c,\ u(S) \leq h(S)\} = 0$ with $h(S) := f(S) - \mu|S| \geq 0$. LP duality supplies α and $\beta \geq 0$ with $\alpha + B_k \geq [\, k \in P \,]$ for every player k, where $B_k = \sum_{S \ni k} \beta_S$, and $\alpha c + \sum_S \beta_S h(S) \leq 0$. Since $c > 0$ and $h \geq 0$, necessarily $\alpha \leq 0$; put $A = -\alpha \geq 0$, so $B_k \geq 1 + A$ on P and $B_k \geq A$ on R.

Step 3 (exact tightness). $x \in \mathrm{Core}(v)$ gives $h(S) \geq x(S) - \mu|S| = (\varrho - \mu)\,|S \cap R|$, so

$$\sum_S \beta_S h(S) \;\geq\; (\varrho - \mu)(B_{r_1} + B_{r_2}) \;\geq\; 2A(\varrho - \mu) \;=\; cA,$$

using $2(\varrho - \mu) = 1 - 5\mu = c$ *exactly*. Step 2 bounds the same sum by cA, so both inequalities are equalities: $B_{r_1} + B_{r_2} = 2A$, and every support set of β is f-tight.

Step 4 (counting). The key lemma now applies to every support set, so $B_{p_1} + B_{p_2} = \sum_S \beta_S |S \cap \{p_1, p_2\}| \leq \sum_S \beta_S |S \cap R| = 2A$, while Step 2 demands $B_{p_1} + B_{p_2} \geq 2 + 2A$. Contradiction. □

Remark 26 (sharpness, and the archived check)**.** *The constant is not inserted; it falls out. Exactly one line of the proof mentions* $\frac{1}{8}$ *— the coalition shape* $U = \{p_i\} \cup R$ *— and at* $\mu = T/8$ *that inequality becomes an equality and the key lemma fails at* $S = \{j, p_{i'}\}$*. This is realized: in the extremal game of Theorem 21 (which lies in the configuration with* $\mu = T/8$*) the* f*-tight sets* $\{1, 2\}$ *and* $\{2, 4\}$ *are precisely of that shape. The proof also consumes strictly less than the original finite check did: neither the hard direction of Theorem 23 nor the facet description appears — only core membership of the two selections, the floor bracketing* $\mu \le \lambda$*, Corollary 30, and the pinning. The original check (*10,974 *exact programs over the* 354 *pinning-dual vertices) remains archived as an independent confirmation.*

Corollary 27 (the pinned solution pair)**.** *In the configuration of Theorem 25, suppose in addition that the starved player is* Lmin*-poor and that some coalition attaining* ν *has three players. Then, up to relabelling,* $\mathrm{Lmin} = \left(\frac{T}{8}, \frac{T}{8}, \frac{T}{8}, \frac{5T}{16}, \frac{5T}{16}\right)$*,* $\mathrm{Lmax} = \left(0, \frac{T}{4}, \frac{T}{4}, \frac{T}{4}, \frac{T}{4}\right)$*, and* $\rho(v) = \frac{1}{2}$ *exactly.*

Proof. Normalize $T = 1$. The Lmax pattern gives $M = 1/4$ directly. Core membership of the Lmax vector gives, for $S \neq \{j\}$, $f(S) \ge \big(|S| - [\, j \in S \,]\big)/4 \ge |S|/8$, and Theorem 25 gives $f(\{j\}) \ge 1/8$, so $\lambda \ge 1/8$. By the M-side of Theorem 23, $\nu = M = 1/4$, so a three-player coalition S_0 attaining ν has $v(S_0) = 3/4$ and $f(N \setminus S_0) = 1/4$ on two players, so $\lambda \le 1/8$; hence $\lambda = 1/8$, and Theorem 23 gives $\mu = 1/8$. With the starved player Lmin-poor, $\mu = 1/8 \le 1/6$ gives $(1 - 3\mu)/2 \ge 1/4 = M$, so the coordinatewise gap telescopes to $\| \mathrm{Lmin} - \mathrm{Lmax} \|_1 = T - 4\mu = T/2$, and the two displayed allocations are the unique ones consistent with the strata, the floor and the totals. □

The first additional hypothesis is automatic: Corollary 30 below proves, in two lines, that the starved player is always Lmin-poor. The second — a three-player coalition attaining ν — holds in every configuration game in the combined harvests (53 of 53) but remains open in general, and is recorded in Section 7.

For the ratio *bound*, as opposed to the pinned pair, both hypotheses can in fact be dispensed with.

Theorem 28 (unconditional configuration bound)**.** *In the configuration of Theorem 25,* $\rho(v) \le \frac{1}{2}$.

Proof, by gap arithmetic. Normalize $T = 1$; the Lmax pattern gives $M = 1/4$ and the $(3, 2)$ pattern gives $\mathrm{Lmin} = (\mu, \mu, \mu, \varrho, \varrho)$ with $\varrho = (1 - 3\mu)/2$. The starved player is Lmin-poor by Corollary 30 below (a two-line consequence of Lemma 29, independent of this theorem), so the coordinatewise gap is $1 - 4\mu$ for $\mu \le 1/6$ and 2μ otherwise, and Theorem 25 gives $\mu \ge 1/8$, so both branches are at most $\frac{1}{2}$. (Historically the Lmin-rich branch, whose gap $3/2 - 6\mu$ would exceed $\frac{1}{2}$ for $\mu < 1/6$, was first closed by a finite check of the same design as Theorem 25 — the rich relabelling of the pinning polyhedron, 31×354 exact programs, all infeasible, with the drop-the-pinning control restoring feasibility. The corollary supersedes that check by showing the branch is vacuous; the check remains archived as an independent confirmation.) □

Two calibrations. First, the floor bound is not removable in favour of soft arguments: from Theorem 23 and core membership of both allocation vectors alone — everything that is game-free — the best obtainable bound on ρ at five players is exactly $\frac{3}{2}$, so every configuration needs a lower bound on the floor of the strength of Theorem 25. Second, the attempted refutation of the ν-triple hypothesis produced, once again, a certificate whose audited model is a genuine exact game *outside* the configuration (a $(1, 4)$-shaped Lmax with minimum $1/30$, ratio $5/12$) — the third certificate-not-realizable instance of this program, and the reason hypotheses are retired here only by finite checks, never by feasible certificates alone.

The configuration hypothesis is essential, not decorative: dropping the $(3,2)$ pinning makes the same programs satisfiable, and their models are genuine exact five-player games with $T-v(N\setminus\{i\}) = T/16$ for some player i — half the bound — but with Lmin = Lmax, hence no separation at all. The cases outside Corollary 27 — the Lmax stratum shapes $(1,2,2)$ and $(1,1,3)$, the $(4,1)$ Lmin boundary, and the Lmin-rich starved player — are closed in the next subsection, which settles the supremum in full.

5.5 The five-player supremum in full

Three localization lemmas come first; all are consequences of Theorem 23 together with core membership of the two allocation vectors, and none needs a computation.

Lemma 29 (anchor localization)**.** *Let v be a singleton-normalized exact five-player game, S_0 a coalition attaining ν and S_* a coalition attaining λ (both exist by Theorem 23). Then*

(a) $|S_0| \geq 2$, and $\mathrm{Lmax}_i = M$ for every $i \in S_0$; consequently the Lmax *top stratum contains at least two players;*

(b) $\mathrm{Lmin}_i = \mu$ for every $i \in S_*$, and $\mathrm{Lmax}(S_*) \leq \mu|S_*|$; consequently $\mu \geq \min_{i\in P} \mathrm{Lmax}_i$, where P is the Lmin *bottom stratum;*

(c) $2M + 3\mu \leq T$.

Proof. (a) If $S_0 = \{i\}$ then $M = \nu = v(\{i\}) = 0$, but $\mathrm{Lmax}(N) = T > 0$ forces $M \geq T/5$. For the second claim, $\mathrm{Lmax}(S_0) \geq v(S_0) = M|S_0|$ by core membership while $\mathrm{Lmax}_i \leq M$ throughout, so Lmax is constant M on S_0. (b) Core membership gives $\mathrm{Lmin}(S_*) \leq f(S_*) = \mu|S_*|$ while $\mathrm{Lmin}_i \geq \mu$ gives the reverse, so Lmin $= \mu$ on S_*; the same cap applied to Lmax gives $\mathrm{Lmax}(S_*) \leq \mu|S_*|$, whence $\mu \geq \mathrm{Lmax}(S_*)/|S_*| \geq \min_{i\in S_*} \mathrm{Lmax}_i \geq \min_{i\in P} \mathrm{Lmax}_i$. (c) By the floor side, $f(N\setminus S_0) \geq \mu(5-|S_0|)$, i.e. $M|S_0| + \mu(5-|S_0|) \leq T$; since $M \geq T/5 \geq \mu$ the left side is nondecreasing in $|S_0|$, and $|S_0| = 2$ is admissible by (a). □

Part (b) sharpens Corollary 24: the minimum ranges over the Lmin-*poor* players only. Combined with the trivial $\mu \leq T/5$ it closes, by two lines, the hypothesis that resisted the configuration analysis.

Corollary 30 (the starved player is Lmin-poor)**.** *In the configuration of Theorem 25 —* Lmax *strata $(1,4)$ with the singleton at zero, at player j — the player j lies in the* Lmin *bottom stratum.*

Proof. The stratum pattern forces $4M = T$. If $j \notin P$ then $\mathrm{Lmax}_i = M = T/4$ for every $i \in P$, so Lemma 29(b) gives $\mu \geq T/4$, contradicting $\mu \leq T/5$. □

In particular the Lmin-rich branch of Theorem 28's proof is vacuous, the first additional hypothesis of Corollary 27 is automatic, and the corresponding finite check (31×354 programs) is superseded by the two lines above.

The second ingredient is an exact certificate form for the two selections, replacing anchors and pinnings by linear identities.

Lemma 31 (staged Farkas certificates)**.** *Let v be a game with nonempty core, ϕ a core point with level sets $B_1 \prec \cdots \prec B_k$ in increasing order of payoff. Then $\phi = \mathrm{Lmin}(v)$ if and only if for every stage t and every $i \in B_t$ there exist multipliers $y_A \geq 0$ supported on proper ϕ-tight coalitions, $\eta \in \mathbb{R}$, $\rho_l \in \mathbb{R}$ for $l \in B_1 \cup \cdots \cup B_{t-1}$ and $\sigma_c \geq 0$ for $c \in B_t$ with, for every player j,*

$$-\sum_{A\ni j} y_A + \eta + [\,j \in B_{<t}\,]\rho_j - [\,j \in B_t\,]\sigma_j \;=\; [\,j = i\,].$$

The mirror statement characterizes Lmax*, with the level sets taken in decreasing order, σ entering with a plus sign, and right-hand side $-[\,j = i\,]$.*

Proof sketch. Leximin is computed stagewise: with the earlier strata pinned, the level μ_t is the largest uniform lower bound on the unpinned players, and B_t collects the players whose maximum subject to that equals μ_t. $\phi = \mathrm{Lmin}(v)$ is equivalent to the family of stage conditions $\max\{x_i : x \in \mathrm{Core}(v),\ x_l = \phi_l\ (l \in B_{<t}),\ x_c \geq \mu_t\ (c \in B_{\geq t})\} = \phi_i$ for $i \in B_t$. Each is a linear program whose feasible set contains ϕ; the constraints active at ϕ are efficiency, the ϕ-tight coalitions, the pinned equalities, and the level bounds on B_t itself. Dualizing over the tangent cone at ϕ and computing the polar cone gives exactly the displayed identity; complementary slackness confines the support of y to tight coalitions and eliminates the bilinear terms $y_A v(A)$, which is what makes the certificate linear in (v, ϕ, y) jointly. The converse direction is the same computation read backwards. The certificate was derived independently twice within the program (once per player and stage as displayed, once per payoff cut) and the two forms verified against each other in both directions on 30 games and 6 non-optimal core points; the mirror was re-derived from scratch and validated on 1050 games (5,250 ceiling certificates, 10,500 counting the floor form, no failures), with negative controls rejecting a false Lmax claim and the core midpoint 261/261 times each. □

With these in place the supremum closes. Recall that a *configuration* is a triple: the Lmin stratum composition, the Lmax stratum composition, and the table of overlaps between the two stratifications. At $n = 5$ there are 2961.

Theorem 32 (the $(4, 1)$ regime)**.** *Every singleton-normalized exact five-player game whose* Lmin *strata form the pattern* $(4, 1)$ *satisfies $\rho(v) \leq \frac{1}{3}$, and the bound is attained.*

Theorem 33 (five-player supremum)**.** *Every singleton-normalized exact five-player game satisfies $\rho(v) \leq \frac{1}{2}$. With Theorem 21, $s(5) = \frac{1}{2}$.*

Proof of both, by configuration-complete finite check. Relabelling players reduces every game to the canonical layout of its configuration, so it suffices to exclude $\rho > \frac{1}{2}$ (resp. $\rho > \frac{1}{3}$ on the $(4, 1)$ class) in each configuration. This was done by two pipelines whose Lmin-optimality components are independently formulated; no broader machinery-independence claim is made. Coverage is machine-audited: every one of the 2961 configurations lands in exactly one bucket, no run errored, and none was left feasible.

The first pipeline is structural: 1953 configurations are impossible by Lemma 29(a) alone (Lmax composition ending in a singleton); 800 are excluded by one game-free exact program each — levels, orderings, totals and the anchor conditions of Lemma 29 against $\rho > \frac{1}{2}$; 100 fall to a finite check with v free, both core memberships, the certificates of Theorem 23, the floor and ceiling stage-one pinnings (dualized as in Theorem 25), and the exact-cone facet inequalities as lazy cuts; and the remaining 108 fall to the same check with the complete multi-stage pinning ladders — every Lmin stratum pinned given the strata below it, every Lmax stratum pinned given the strata above it, the reduced-game caps encoded exactly as minima over preimages. All programs are infeasible in exact rational arithmetic with no indeterminate verdicts.

The second pipeline re-proves the same verdicts from a minimal hypothesis basis: singleton normalization and scale; core membership of both vectors; exactness imposed *existentially* (for each proper coalition S, an attaining core point w^S with $w^S(S) = v(S)$ — the definition of exactness, linear, with no appeal to the facet description); and Lmin/Lmax imposed through Lemma 31 and its mirror. All 2961 programs are again infeasible — so the theorems depend neither on Theorem 23 nor on the facet catalogue. The two pipelines agree configuration by configuration.

Soundness controls, both pipelines: dropping the pinning ladders (resp. certificates) restores feasibility wherever the configuration is possible at all; the hypothesis sets hold, in exact arithmetic, on 1050 profiled random exact games spanning 21 configurations — the first pipeline's 2874 stage certificates in one run, the second pipeline's 10,500 Farkas certificates in another, no failures in either; and on all eight configurations for which a game above some target ratio is known, the same programs run at that target return feasible models that *audit end-to-end* as genuine exact games in the assumed configuration — including the extremal configuration at ratio exactly $\frac{1}{2}$ and the $(4, 1)$ regime at exactly $\frac{1}{3}$. The configurations declared possible include every configuration any harvest has produced. Attainment in Theorem 32 is by explicit games (111 harvested at exactly $\frac{1}{3}$). All drivers, logs and the coverage audit are archived with the verifiers. □

The two-regime picture is therefore a theorem: Lmin strata $(4, 1)$ force $\rho \leq \frac{1}{3}$ while $(3, 2)$ allows $\frac{1}{2}$, both attained. One calibration deserves record: under the single-stage hypothesis set the final 108 configurations are all *satisfiable*, and every one of the 108 models audits as a genuine exact game — in the wrong configuration. The complete ladders are what make the encoding decisive rather than merely stronger; this is the fourth, and largest, instance of the certificate-not-realizable phenomenon in this program, and the reason every feasible model here is audited before it is believed.

6 Obstructions to an upper bound

6.1 The exact six-player bound

Conceptually, this is a theorem about a core polytope inside the normalized simplex. Singleton exactness makes that polytope touch every coordinate facet, through six witness allocations. If two canonical Lorenz selections were farther than 2/3, their coordinatewise comparison would have one of fifteen sign patterns. In each pattern, every infinitesimal exchange that improves one selection must be stopped by a tight coalition on which the exchange has negative sum; the opposite selection and the six facet witnesses impose seven necessary inequalities on that blocker. The finite part proves that no such blockers can coexist. Thus the computation is a fifteen-case convex-geometric exhaustion of necessary exchange obstructions, not an enumeration of games or core vertices.

The first nontrivial fixed dimension can now be closed without a generator normal form or a support-table census.

Proof architecture. There are three dependencies and only three. A human reduction maps any strict counterexample to one of fifteen sign-labelled systems of necessary exchange blockers. A finite linear-arithmetic lemma proves those systems infeasible, with independently checked proof objects. Finally, the explicit game $G(2, 3)$ supplies the matching lower bound. The earlier generator census, equality census, continuity arguments, and support tables are historical discovery routes and are not used in this proof.

After orbit reduction, the nineteen-schema formulation of the three large sign cases uses only the following hand-checkable exchange calculus. The original minimized labelled menus contain additional direction shapes; they are proof-checked in the fifteen-case formulation but are not being classified by this lemma. Naming the reduced templates separately isolates what is conceptual from the later blocker-coexistence certificates.

Lemma 34 (Three exchange templates)**.** *Let $y \in \mathbb{R}^6$ and let $h(N) = 0$.*

(i) For $h = -e_a + e_r$, h is Lmin-admissible exactly when $y_r < y_a$; it has 16 negative nonempty proper cuts. The Lmax condition is the same inequality $y_a > y_r$, read at the highest affected level.

(ii) For $h = -e_a - e_b + e_r + e_s$, h is Lmin-admissible exactly when $\min(y_r, y_s) < \min(y_a, y_b)$; it has 20 negative nonempty proper cuts. For Lmax, the condition is $\max(y_a, y_b) > \max(y_r, y_s)$.

(iii) For $h = e_p + 2e_{p'} - e_{m_1} - e_{m_2} - e_{m_3}$, h is Lmax-admissible exactly when $\max_j y_{m_j} > \max(y_p, y_{p'})$; it has 24 negative nonempty proper cuts. Negating h gives the Lmin dual.

Ties make the displayed strict activation fail; they introduce no additional case.

Proof. Admissibility asks whether the first affected payoff stratum contains a coefficient of the improving sign and none of the opposite sign. Reading the lowest affected stratum for Lmin and the highest for Lmax gives the stated minimum and maximum tests. For (i), negativity means $a \in S, r \notin S$, with the other four players free. For (ii), if d donors and u recipients belong to S, negativity is $d > u$; summing the cases $(d, u) = (1, 0), (2, 0), (2, 1)$ gives $8 + 4 + 8 = 20$. In (iii), negativity means $|S \cap \{m_1, m_2, m_3\}| > \mathbf{1}_{p \in S} + 2\mathbf{1}_{p' \in S}$; the unused sixth player is optional, and the three positive-side loads contribute $14 + 8 + 2 = 24$ cuts. Negation proves the dual statements. □

Theorem 35 (Six-player bound)**.** *Every singleton-normalized exact six-player game satisfies*

$$\| \operatorname{Lmin}(v) - \operatorname{Lmax}(v) \|_1 \leq \frac{2}{3} v(N).$$

The bound is attained by $G(2, 3)$, and hence $s(6) = 2/3$.

Proof. Scale $T = v(N)$ to one and write $x = \operatorname{Lmin}(v)$, $z = \operatorname{Lmax}(v)$. Put

$$P = \{i : x_i > z_i\}, \qquad M = \{i : z_i > x_i\}, \qquad E = \{i : x_i = z_i\}.$$

If $x \neq z$, both P and M are nonempty. Up to a common relabelling there are therefore fifteen triples (p, m, e) with $p, m \geq 1$ and $p + m + e = 6$, and

$$\|x - z\|_1 = 2 \sum_{i \in P} (x_i - z_i). \tag{28}$$

Exactness at each singleton supplies a core point q^i with $q^i_i = 0$. For a point y and direction h, define the first order-crossing time

$$\Delta(y, h) = \min \left\{ \frac{y_j - y_i}{h_i - h_j} : y_i < y_j,\ h_i > h_j \right\},$$

with the empty minimum equal to $+\infty$. Every $0 < \varepsilon < \Delta(y, h)$ preserves all strict orders between distinct old payoff strata.

Three elementary selection anchors will be used. First, z cannot have a unique top player i: along the core segment toward q^i, the direction $q^i - z$ is negative at that unique top because $q^i_i = 0 < z_i$. Taking $0 < \varepsilon < \min\{1, \Delta(z, q^i - z)\}$ lowers the first entry of the decreasing rearrangement. Second, some z-top player has $x_i \geq z_i$, since otherwise the direction $x - z$ is negative throughout the old top stratum and the same explicit bound lowers the first changed decreasing-lex entry. Dually, some x-bottom player has $z_i \leq x_i$, using the direction $z - x$. Convexity keeps all three segments in the core; the crossing bound handles tied strata without an informal limiting argument.

Now let $h(N) = 0$. On the Lmin side call h admissible if its lowest affected payoff stratum has a positive coefficient and no negative coefficient. On the Lmax side call it admissible if its

highest affected stratum has a negative coefficient and no positive coefficient. Such a direction strictly improves the relevant sorted lexicographic vector for every positive step below $\Delta(y,h)$. If all proper tight coalitions had nonnegative derivative, set

$$\varepsilon_* = \min\left(\{1, \Delta(y,h)\} \cup \left\{\frac{y(S)-v(S)}{-h(S)} : \varnothing \neq S \subsetneq N,\ y(S) > v(S),\ h(S) < 0\right\}\right).$$

Every finite member is positive and the set contains 1, so $\varepsilon_* > 0$. Any $0 < \varepsilon < \varepsilon_*$ preserves both the payoff-stratum order and every core inequality, while $h(N) = 0$ preserves efficiency. This would give a core improvement, a contradiction. Optimality therefore forces a proper tight coalition S with $h(S) < 0$. Core membership of the opposite selection and the six witnesses gives, respectively,

$$z(S) \geq x(S), \quad q^i(S) \geq x(S)\ (i \in N), \tag{29}$$

or

$$x(S) \geq z(S), \quad q^i(S) \geq z(S)\ (i \in N). \tag{30}$$

As a finite differential control, the pivot implementation of admissibility was compared with a separately written first-affected-stratum criterion on all 4,683 labelled weak payoff orders, all 1,610 bound-two directions, and both selections: 15,079,260 comparisons with no mismatch.

For each sign triple, impose nonnegativity and efficiency of x, z, q^i, the strict signs, (28) with left side $> 2/3$, the three anchors, and the necessary blocker disjunction (29) or (30) for a finite menu of admissible integer directions. Thirteen cases use only efficiency-preserving directions with coefficients in $\{-1, 0, 1\}$; $(2,3,1)$ and $(3,3,0)$ use coefficients in $[-2, 2]$. After deletion of redundant direction clauses the fifteen menu sizes are

$$1, 1, 1, 1, 0, 2, 5, 14, 8, 2, 52, 71, 2, 90, 2, \tag{31}$$

in lexicographic order of (p, m, e). Every resulting exact QF_LRA formula is infeasible. Z3 and cvc5 agree on every formula; cvc5 emits CPC with proof pedantry enabled and trust disabled, and Ethos checks all fifteen proofs as `correct`. The formulas are relaxations: they retain only necessary consequences of tightness, so their infeasibility excludes every exact-game counterexample. The source-to-formula regeneration, hashes, proof controls, and all retained directions are in the repository under `verifiers/s6/strict_exchange_*`.

The dependency structure is simpler than the discovery history. First, the game-theoretic lemmas above map any genuine counterexample into the finite relaxation. Second, the retained exchange menus give an exhaustive relay of admissible directions and necessary blockers. Third, exact rational identities close every arithmetic leaf. Solver searches were used to discover and minimize these objects but are not evidence for the last layer. Nine of the eleven localized packet rows have independently specified hand-checkable relays; the two largest have complete solver-free arithmetic conditional on frozen relay/use maps. External validation of the first layer and independent reconstruction of those two maps remain distinct from rerunning the solvers.

The two bound-two cases were additionally reconstructed by a separate encoder that imports none of the production encoder, lex predicate, blocker builder, direction generator, minimizer, or SMT serializer. Its base systems are satisfiable, its full systems are infeasible in both solvers, and its fresh trust-disabled CPC proofs check in Ethos. A third formulation imports neither encoder nor Z3: a standard-library generator writes SMT-LIB directly and expresses activation as strict separation of the positive and negative payoff supports. All 458 assertion comparisons with both prior formulations are equivalent; both third-formulation bases are satisfiable and both full systems are infeasible. These controls reduce correlated encoding risk but do not replace

external review of the human implication. At the orbit level, the three largest menus admit a further unguarded reduction: their 213 labeled directions and 44 sign-role orbits shrink to 136 directions in 19 orbits. By Lemma 34, the $(3, 2, 1)$ and $(4, 2, 0)$ cases then use only one-for-one and two-for-two exchanges; the balanced $(3, 3, 0)$ case adds one asymmetric coefficient-two shape and its reverse. Their activation side is completely described by the same 19 role-level order-statistic tests: compare the lowest recipient with the lowest donor for Lmin, or the highest donor with the highest recipient for Lmax. Their 2,512 eligible negative cuts use only three unoriented templates, with respectively 16, 20, and 24 cuts for one-for-one, two-for-two, and asymmetric coefficient-two exchange. The remaining finite content is therefore blocker coexistence, not lexicographic activation or cut enumeration. All three reduced formulas have separate trust-disabled CPC proofs checked by Ethos, and deleting any selected orbit makes the corresponding relaxation feasible. These controls reduce all three large cases using only the two cross-selection anchors; the no-unique-Lmax-top anchor is dispensable there. In $(4, 2, 0)$ they further reduce each blocker alternative to the opposite-allocation comparison and the four singleton-witness comparisons indexed by P; both M-indexed comparison families are dispensable, while each of the five retained sources is irredundant in the relaxation. Anchor localization is sharper: if the x-bottom and z-top anchors $a, b \in P$ coincided, then $x_m \geq x_a > z_a \geq z_m > x_m$ for $m \in M$. After relabelling the distinct anchors, a separate 18-real certificate uses nineteen directions in ten residual orbits and only the one exactness witness q^b. These controls reduce implementation correlation and make the finite anatomy readable. In that localized case two exchanges are automatic. Their negative blockers form a pair $\{b, 4\} \cup A, \{b, 5\} \cup B$ with $A, B \subseteq P \setminus \{b\}$. Of the 64 labelled pairs, 32 are base-infeasible and the other 32 form eleven classes under the residual symmetry; every class closes with a subset-minimal menu of at most seven remaining exchanges. Adding the two blocker inequalities also gives a short aggregate packet inequality by hand. This eleven-row decomposition has nine hand-checkable rows. One is a two-exchange blocker relay whose four terminal blocker pairs force the distance back to at most $1/3$. In the second packet orbit the single exchange $-e_1 - e_3 + e_4 + e_5$ is forced to improve Lmin, while every one of its twenty eligible blockers violates the opposite-allocation comparison. The third is a depth-one three-leaf star: the root activation, exact three-blocker list, and all child activations have structural inequalities. Its terminal Lmax leaf has sixteen possible complement cuts and its two terminal Lmin leaves reduce to four base cuts. Twenty explicit nonnegative weighted identities, checked by pure rational coefficient arithmetic without a solver, show that every such cut forces the distance to at most $1/3$. The fourth is a unique three-exchange chain. Its first twenty cuts reduce structurally to seven; six source-specific identities leave the unique first blocker, fifteen leave the unique second blocker, and sixteen exclude every terminal cut. The fifth is the depth-three class: its apparent large tree reduces to eight exact difference-cone cut censuses, sixteen source-specific blocker filters, and nine activation identities. Exact Fourier–Motzkin elimination independently checks the cut table, and pure rational arithmetic checks all 25 identities. The sixth class closes after the exhaustive split $x_3 \leq x_4$ versus $x_3 > x_4$. Its five-leaf relay reduces to ten exact difference-cone cut censuses and 71 certificate uses. Deduplication leaves 43 blocker identities, seven activation identities, and two terminal-anchor identities; the terminal leaves share the threshold contradiction $x_1 > 1/3$ versus $x_1 \leq 1/3$. A solver-free exact-rational checker reconstructs every premise at every use site and rejects coefficient, use-site, candidate, use-map, and strict-target mutations. The seventh class closes after the same root order split and one nested comparison. Its nine leaves reduce to eighteen exact cut censuses and 108 identities checked at 122 uses. All terminal leaves force the distance directly to at most $1/3$; 103 of its 105 blocker exclusions use only the single witness q^1, while two weak-branch exceptions use only the opposite selection. These first seven hand-checkable rows cover twenty-four labelled packets. The eighth class is the singleton packet orbit. After the universal root split, its fifteen-leaf relay reduces to twenty-one exact difference-cone censuses and 85

blocker exclusions. The disjunctive two-donor activation is equivalent to comparing the minimum donor level with the minimum recipient level, reducing its failure expansion to four conjunctions and 75 cases over the tree. Deduplication leaves 65 blocker and 44 activation identities checked at all 160 uses by a solver-free rational checker. The ninth class has an asymmetric packet sandwich and an eleven-leaf relay. The same minimum-level lemma and its maximum-level dual reduce its activation side to 72 cases; 21 exact cut censuses and 108 identities close all 132 uses. Thus the nine hand-checkable rows cover twenty-seven labelled packets. The largest remaining row has a symmetric packet satisfying $d_0 + d_1 \le \delta_4, \delta_5 \le d_2 + d_3$; the anchor chains $z_0 < x_0 \le x_4 < z_4 \le z_1$ and $x_0 \le x_4 < z_4 \le z_1 < x_1$ make both one-for-one exchanges automatic. For the Lmin exchange $e_0 - e_1$, the same sandwich reduces all sixteen eligible negative cuts to ten possible opposite blockers, uniformly at all ten occurrences in the frozen tree. Exact coefficient arithmetic checks the sandwich and cut partition without a solver or production import. Three orbit identities further exclude seven of the thirty formal pairs between the three possible preceding blockers and the ten packet candidates. The root split is hand-classified as well: its weak and strict twenty-cut censuses reduce to three and four blockers, respectively, using one seven-term witness identity beyond signs. Across all 90 direction nodes, 1,736 eligible cuts split exactly into 1,464 pure difference-cone exclusions, 34 packet/ancestor-context exclusions, 150 target-witness exclusions, and 88 relay branches. The 184 exclusions have sparse incidence over-covers and reduce to 107 exact rational identities. The 285 elementary activation failures split into 195 exact allocation contradictions and 90 unit-weight witness over-covers; 72 exact rational identities settle all of them. Solver-free checkers verify every identity and the exhaustive partitions. Thus no arithmetic leaf in the 51-terminal tree remains machine-only, conditional on its frozen relay/use maps. Its 24 labelled cut menus collapse to fifteen rich-player-parametric templates, and a path-local grammar generates all 88 edges and 51 leaves. Forty-one path-free activation identities cover 253 uses; only 32 uses require an ancestor comparison, in three relay families. The rich-player swap reduces the 72 identities to 47 affine motifs, twenty of them path-dependent. This is still not a tenth independently derived hand proof: the grammar and the 32 residual placements require external reconstruction. The other remaining row has an asymmetric packet sandwich $d_0 + d_1 \le \delta_4 \le d_3$ and $d_0 + d_1 + d_2 \le \delta_5 \le d_2 + d_3$; it reduces both twenty-cut roots to two blockers using signs alone. Across its 49 direction nodes there are 924 eligible negative-cut uses. Exact Fourier–Motzkin elimination removes 711 in the pure difference cone and leaves 213 candidates. The ten distinct candidate sets form seven direction menus plus one binary refinement: below the weak-root h_{11} blocker 124, the h_7, h_{10}, h_{15} menus each gain one explicit cut. Seven path-local transition rules then generate all 47 relay edges: they depend only on the weak/strict regime, the preceding h_{10} cut, and the same $h_{11} : 124$ bit. Eleven candidates are excluded by the already active packet/ancestor witness context, 155 by the target singleton witness, and 47 are exactly the relay branches. The eleven intermediate exclusions have nine solver-free rational identities. All 155 target-witness exclusions have incidence over-covers of support at most five; 150 use unit coefficients and five require coefficient two. This is a complete human explanation of the witness-side upper bounds, but the allocation-side strict reductions have also been exactified: 53 rational identities, checked solver-free at all 155 uses, show that the corresponding lower sums strictly exceed their incidence bounds. Of the uses, 118 need no path comparison, 33 use ancestors only, and four also use the target comparison. An independent exact-arithmetic checker verifies that the $11 + 155 + 47$ uses exhaust all 213 candidates, so every terminal no-blocker closure is arithmetic rather than solver-only. The coverage maps remain relay-derived. The same incidence geometry controls activation: anchor/order chains remove 84 of 124 elementary failure cases, four more are allocation-infeasible, and each of the other 36 has a packet/ancestor over-cover of support at most four. The activation incidence bounds are human; 26 solver-free rational identities prove the 36 allocation lower bounds, and the four direct cases share one more identity. Thus no class-6

arithmetic leaf remains machine-only, conditional on the frozen relay/use maps. Of the full 40-use activation map, 19 uses are path-free and the 21 path-dependent uses split into three relay families; the 27 identities reduce to 25 affine motifs after source-depth aliases are forgotten. Those maps remain encoding-shared and require external human-to-machine review. Both remaining rows also close after imposing the same root split $x_3 \le x_4$ versus $x_3 > x_4$. Indeed, that single split gives complete forced-exchange relay trees for all eleven rows; class 10 alone needs the nested x_1–x_2 split. The last two trees still rely on the parent encoding rather than separate exact hand certificates. This does not replace the human obligation to verify the counterexample-to-formula implication.

The single-witness filters have a common exact geometry. For any fixed coalition menu, the map $q \mapsto (q(C))_C$ sends $\{q \ge 0 : q(N) = 1, q_b = 0\}$ onto the convex hull of the non-anchor incidence columns. Hence a nonnegative weighted sum of witness comparisons is bounded exactly by its maximum non-anchor incidence load, attained at a unit vector. The six witnesses are otherwise independent, so after choosing one blocker for each active exchange this projection eliminates all witness coordinates exactly; their six anchor-specific support bounds collapse to the second-largest player incidence load. This support system has an exact finite reduction. For anchor i and a proposed maximizing non-anchor player k, partition the nonnegative weight orthant by the cones

$$K_{i,k} = \{\lambda \ge 0 : \lambda \cdot (a^k - a^\ell) \ge 0 \text{ for every } \ell \notin \{i,k\}\}.$$

On $K_{i,k}$ the support bound is linear, so its extreme rays give a complete rational inequality system. A ray with s positive coordinates requires $s-1$ active player-comparison hyperplanes; because there are only $n-2$, $s \le n-1$. Thus every six-player witness inequality uses at most five blocker requirements, and a five-coalition incidence circuit shows this bound is sharp. This constructive ray statement strengthens the corresponding pointwise Helly bound and gives an exact facet-enumeration algorithm. The remaining system is a finite cut-transversal problem coupled to this symmetric incidence support function and the opposite-selection signs. The class-5 crossing sums are sparse incidence over-covers of this form. This standard support-function identity supplies an LP/facet algorithm. Applied exhaustively to class 6, it gives the 155 target-witness supports and eleven context supports just described; these couple to 53 and nine exact allocation identities, respectively. The exact cut census identifies the remaining 47 candidates with the relay branches and settles every terminal no-blocker closure. This arithmetic exhaustion does not independently validate the relay/use maps or the counterexample-to-formula implication.

Thus the upper bound holds. The family formula gives $\rho(G(2,3)) = 2(2(3-1)-1)/(3(2+1)) = 2/3$, proving equality. □

6.2 No constant below 2

Theorem 17 already rules out one shape of answer. Any bound of the form $\|\operatorname{Lmin}(v) - \operatorname{Lmax}(v)\|_1 \le c \cdot v(N)$ with a constant $c < 2$, valid for all n, is false: choose n even with $2 - 8/n > c$. So the only possible clean upper bound is dimension-dependent, and the right target is the finite-n statement

$$s_2(n) = 2 - \frac{8n}{n^2 - (n \bmod 2)} \qquad (n \ge 6),$$

now proved for the full exact class at $n = 6$ by Theorem 35, and confirmed by complete two-generator enumeration at $n = 7, 8$, together with the question of whether $s(n) = s_2(n)$ in larger dimensions.

6.3 No bound linear in the leximin floor

A second shape is also excluded, and this one was a live candidate. The inequality

$$\|\operatorname{Lmin}(v) - \operatorname{Lmax}(v)\|_1 \;\le\; 4(n-4)\,\mu(v), \qquad \mu(v) = \min_i \operatorname{Lmin}(v)_i,$$

holds with no violations in 900 fresh random five-player games and is tight both on the $n = 5$ extremal class and on $G(3,4)$ at $n = 8$, where both sides equal 12. It fails from $n = 9$ on:

game	n	$\|\operatorname{Lmin} - \operatorname{Lmax}\|_1$	$4(n-4)\mu$
$G(4,4)$	9	88/5	16
$G(3,5)$	9	33/2	15
$G(4,5)$	10	24	96/5
$G(5,5)$	11	95/3	70/3

The reason is structural rather than accidental: in $G(p,k)$ one has $\mu = p/(p+1) < 1$, bounded, while the gap grows like $2T$. No bound of the form $C(n)\mu(v)$ can track the divergence, whatever C is.

6.4 Non-monotonicity under majorization

The most natural route to an upper bound on the two-generator class is to argue that a uniform h-block is without loss of generality — the extremal family $G(p,k)$ has one — and then to optimize over g alone using the floor formula of Conjecture 8. That route is closed.

Define a *smoothing step* as a Robin Hood transfer of one unit from a larger to a smaller h-coordinate, so that h moves down in the majorization order, toward uniform. The premise that a smoothing step never decreases ρ is false, and not only for small blocks:

$\lvert B\rvert$	2	3	4	5
smoothing steps tested	40	76	84	84
violations	18	27	11	21
rate	45%	36%	13%	25%

In particular it fails for $|B| \ge 4$, so the block-size threshold one might guess from the $n = 5$ anomaly does not exist. An explicit verified counterexample at $|B| = 4$, with $n = 7$ and $T = 10$, both allocations confirmed in the core:

$$g = (0,5,0,3,2,0,0), \qquad h = (1,0,2,0,0,5,2),$$

$$\operatorname{Lmin} = (1,2,1,1,1,3,1), \qquad \operatorname{Lmax} = \left(\tfrac{5}{6}, \tfrac{5}{2}, \tfrac{5}{6}, \tfrac{4}{3}, \tfrac{7}{6}, \tfrac{5}{2}, \tfrac{5}{6}\right), \qquad \rho = \tfrac{1}{5}.$$

Smoothing the h-block $(5,2) \mapsto (4,3)$, i.e. replacing h by $(1,0,2,0,0,4,3)$, gives

$$\operatorname{Lmin} = \operatorname{Lmax} = (1,2,1,1,1,2,2), \qquad \rho = 0.$$

So a single Robin Hood step inside one generator collapses the ratio from 1/5 to 0, and ρ is not monotone under majorization of h.

The two facts sit together uncomfortably and both are true: smoothing makes a game more symmetric and tends to *collapse* the two selections onto each other, yet the extremal family has a perfectly uniform h. The dependence is therefore genuinely non-monotone, and any upper-bound proof must handle general h directly rather than reducing to the uniform case.

7 What the results do and do not establish

Proved. Lemma 1 (lower-envelope generation and its converse); Lemma 3 (ceiling and overlap identity); Lemma 4 (support disjointness); Theorem 6 (two-generator core duality, with identity (1) valid without disjointness); Theorem 7 ($\mu \le \lambda$); Proposition 9 (the cover inequality (2) for balanced covers); the validity half of Observation 10; Theorem 13 with Lemma 14 (closed forms for $G(p,k)$, including membership for all $p \ge 1$, $k \ge 2$ and the exact list of tight coalitions); Corollary 15 and Remark 16 (the deficit identity, twice); Theorem 17 ($s(n) \ge 2 - 8n/(n^2 - (n \bmod 2))$ and $\sup_n s(n) = 2$); Lemma 20 (core closing); Proposition 22 (the covering obstruction); Theorem 21 ($\rho = 1/2$ at $n = 5$, by hand). The failure of the $4(n-4)\mu$ bound from $n = 9$, and the non-monotonicity of ρ under majorization of h, are proved by the explicit games displayed in Section 6: each is a finite exact-rational computation on a named game, and each is a refutation, so a single instance suffices.

Proved by exhaustive finite check. Theorem 23 ($\mu = \lambda$ and $M = \nu$ for every exact five-player game; 1142 symmetry classes, each infeasible in exact rational arithmetic), whose M-side Corollary 27 consumes; and Theorems 33 and 32 ($s(5) = 1/2$ and the $(4,1)$ bound $1/3$; 2961 configurations, two pipelines with independently formulated Lmin-optimality components, machine-audited coverage); and Theorem 35 ($s(6) = 2/3$; fifteen table-free sign-count relaxations, independently checked exact CPC proofs). Theorem 25 is now a *hand* result (its one-page proof replaced the original 10,974-program check, which remains archived as confirmation); Lemma 29 and Corollary 30 are hand results; Lemma 31 is a hand result derived independently twice within the program. The checks are proof dependencies; the corollary's remaining additional hypothesis (the ν-attaining triple) is open in general and noted under (4) below.

Enumerated, not proved. The entries of Table 2 are maxima over the stated integer ranges, hence complete only for rational shapes whose primitive integer representative has total in that range; they are lower bounds for $s_2(n)$, not its value. The strict failure of equality in (3) is established by the displayed six-player game, but the claim that this is the *smallest* such failure rests on the enumeration. The tightness rates for (3) and for Conjecture 8 are sample statistics over 586 random games. The structural findings about the $n = 5$ extremal class in Section 5.3 — unique solution pair up to permutation, $(3,2)$ and $(1,4)$ stratum compositions, $|\operatorname{supp}(\mathrm{Lmax})| = 4$, the refutation of the seesaw hypothesis — are computational, over 360 profiled games; of these only the two-regime split is now proved (Theorem 32), the rest remaining sample statistics. The local-optimality certificates from seeded c-space climbing at $n = 5, 6, 7$ are numerical searches over the full exact class; the unseeded variant is under-powered at $n \ge 6$ and certifies nothing.

Conjectured. Four problems remain open.

(1) **Conjecture 8**, equality in the leximin floor formula. Reduced by linear-programming duality to the fractional-cover inequality (2); proved for balanced covers (Proposition 9); the over-covering case is open in general. At five players Theorem 23 *proves* the equality, for every exact game and not only two-generator ones. Verified in 586/586 random two-generator games at $n = 4, \ldots, 7$.

(2) **An upper bound for $n \ge 7$.** Theorem 17 forces any true bound to be asymptotically 2, so every bound $c \cdot v(N)$ with $c < 2$ is false for large n. The right target is $s_2(n) = 2 - 8n/(n^2 - (n \bmod 2))$ for $n \ge 7$, confirmed by complete enumeration at $n = 7, 8$ only. The six-player case is now proved: $s(6) = s_2(6) = 2/3$ (Theorem 35). Section 6 closes

the majorization route and the $C(n)\mu$ route. $n = 5$ is settled (Theorem 33); larger fixed dimensions remain open.

(3) **Is the two-generator class extremal?** Adding a generator lowers the envelope and grows the core, so this is not automatic. All evidence is negative for a counterexample: zero improvements from about 200 third generators at each of $n = 5, 6, 7, 8$, and no improvement from seeded c-space climbing over the full exact class at $n = 5, 6, 7$. Open globally; settled affirmatively at $n = 5, 6$, where the two-generator lower bounds equal $s(5)$ and $s(6)$.

(4) **A structural explanation of the $n = 5$ versus $n \geq 6$ regime change.** The question now has a sharp two-part form, replacing the earlier "why five". *Part one, answered*: the starved family generalizing the extremal configuration (four players at $T/4$, a poor stratum of $n - 2$ absorbing the zeros) exists at *every* $n \geq 5$ with ratio exactly $(n - 4)/(n - 3)$ — now a theorem, by a six-row closed-form construction whose hand proof rests on two coalition values (the identity $T - (n - 3)\mu = 3T/4 = 3M$ lets one cap pin the poorer side from below and fix the ceiling from above, uniformly in n; archived with machine guards at $n = 5..9$). At $n = 5$ the member realizes the extremal configuration and solution pair of Theorem 21 at ratio $\frac{1}{2}$ — though as a *second, non-isomorphic* game (no relabelling aligns the two; the extremal configuration is attained by at least two essentially different games) — so the family is tight there; its value *exceeds* the two-generator bound at $n = 5$, *ties* it at $n = 6$, and undershoots thereafter (it saturates at 1 while s_2 climbs to 2, so it is dominated for large n and says nothing about the supremum beyond $n = 5$); the regime change is the handover between champion families. The crossover test is now resolved by Theorem 35: both families attain the common value $s(6) = 2/3$. *Part two, open*: the floor side of Theorem 23 reduces (archived, with hand lemmas) to a single inequality — for every ordered pair (p, q) of players, some core point satisfies $2y_p + y_q \geq 3\lambda$ — needed only in the regime $\min_{S \subsetneq N} f(S)/|S| \leq T/5$ (outside it the equality $\mu = \lambda$ holds for every game with a nonempty core), and sharp: the margin vanishes on real games. Any proof must engage the dual region in which the efficiency multiplier is negative (active on about 1% of sampled instances), cannot factor through contraction to fewer players, and cannot proceed by saturating the nested pair of caps (both routes are refuted in the archive); and it must consume *exactness*: a balanced non-exact five-player game exists, in the relevant regime, with $\mu = \frac{1}{18} < \frac{1}{12} = \lambda$ (archived and independently re-verified), so core membership alone provably does not suffice. Remaining residues: the M-side of Theorem 23 by hand, a hand distillation of the 2961-configuration check of Theorem 33 (archived progress: 95.5% of configurations fall to twelve hand lemmas), and Corollary 27's ν-attaining-triple hypothesis.

A reduced formulation of problem (2). There is a dimension-free, exactly checkable reduction that narrows the second problem without solving it. Fix one complete staged tight-coalition support $\mathcal{C}_x$ for $x = \mathrm{Lmin}(v)$, one sparse first-stage over-cover support $\mathcal{C}_z$ for $z = \mathrm{Lmax}(v)$, and a singleton exactness witness $w^r \in \mathrm{Core}(v)$ with $w^r_r = 0$. It is sufficient to retain

$$x(S) \leq z(S), w^r(S) \quad (S \in \mathcal{C}_x), \qquad z(S) \leq x(S), w^r(S) \quad (S \in \mathcal{C}_x \cap \mathcal{C}_z),$$

together with the common totals, nonnegativity, the two ordered strata and the actual hybrid sign cut. Indeed the actual triple (x, z, w^r) is feasible in this relaxation, so a lower bound on its hybrid-overlap objective is a lower bound for the game. On a common coalition the displayed rows reduce to $x(S) = z(S) \leq w^r(S)$: the sparse leximax-min certificate glues two rooted tangents only where it meets the complete leximin packet. The over-cover may be chosen with at most n coalitions, and affine Farkas duality gives a support-minimal target certificate with at most $3n - 3$ defining

rows. For a fixed labelled branch, validity, equality vertices and the relative-facet test are finite exact rational computations; projection to a facet of the full exact-game cone still requires the indecomposable semi-balanced test of Studený and Kratochvíl (2022).

This reduction passes exact rational tests at all maximum-overlap roots of the 374 archived six-player pairs (703 rooted programs), as well as five hostile fixed branches and both sharp seven-player orientations. Those are falsification controls, not a proof for any unenumerated domain. An exact six-player negative control drops from 3/4 to 5/8 when the common-hinge rows are removed entirely, so a leximin-packet-only theorem would be false. The open dimension-free step is to prove that some maximum-overlap root and some sparse over-cover always make this common-hinge relaxation attain at least $n/\lfloor n^2/4 \rfloor$, or to produce an exact rational branch that refutes that statement.

One sharper scalar bridge suggested by this reduction is now known to be false. Minimize $\theta_+ = \max_i \alpha_i$ over normalized first-stage ceiling over-covers $\sum_S \lambda_S \mathbf{1}_S = \eta \mathbf{1}_N + \alpha$, and put

$$a = \min\{1/\theta_+, \lfloor n/2 \rfloor\}, \qquad u = a\alpha.$$

Then $0 \le u \le 1$, $u(N) = a$, and scaling the incidence identity gives

$$a \sum_S \lambda_S \mathbf{1}_S + (\mathbf{1}_N - u) = (1 + a\eta)\mathbf{1}_N.$$

If $\epsilon = \sum_S \lambda_S [x(S) - z(S)] \ge 0$, evaluation at the two selections gives $u \cdot x = a(M + \epsilon)$ and $u \cdot z = aM$, with complementary totals on $\mathbf{1}_N - u$. The proposed conjecture was

$$\frac{O(x,z)}{T} \;\ge\; \frac{1}{a} + \frac{1}{n-a}. \tag{CEB}$$

Condition (CEB) would imply the target because $a(n-a) \le \lfloor n^2/4 \rfloor$, is exact on both balanced orientations of the displayed family, and has the correct five/six boundary. Exact archived checks are falsification evidence only: (CEB) passes all 374 stored six-player pairs and the named seven-player controls, while failing on 12 of 124 distinct stored five-player pairs. Those controls missed an exact seven-player counterexample. For

$$g = (12/17, 4/17, 0, 1/17, 0, 0, 0), \qquad h = (0, 0, 6/23, 0, 6/23, 10/23, 1/23),$$

put $v(S) = \min\{g(S), h(S)\}$. The generators have common total one, are core points, attain every coalition worth between them, and have coordinatewise minimum zero; hence v is singleton-normalized and exact. Exact staged certificates give

$$x = (227/782, 48/391, 48/391, 1/34, 48/391, 13/46, 1/34), \quad z = (91, 12, 90, 12, 90, 91, 5)/391.$$

Exact primal and dual incidence witnesses give $\theta_+ = 3/5$ and $a = 5/3$, but

$$O(x,z) = \frac{613}{782} < \frac{63}{80} = \frac{1}{a} + \frac{1}{7-a}$$

with margin $-113/31280$. Thus (CEB) is refuted. The example does not challenge the main target: $613/782 - 7/12 = 941/4692 > 0$.

The cap in (CEB) is not an exogenous parameter. If $\mathcal{T}_z$ is the family of proper coalitions tight at z and H is its top payoff block, ordinary rational LP duality gives the human formula

$$\theta_+ = \max t: \quad y(S) \ge 0 \ (S \in \mathcal{T}_z), \quad y(N) \le 0, \quad t + y_i \le q_i \ (i \in H), \quad q \ge 0, \quad \sum_{i \in H} q_i \le 1. \ \text{(DD)}$$

Formula (DD) remains valid: θ_+ depends only on the labelled top block and its complete tight-coalition incidence hypergraph; adding a tight coalition can only lower it. This explains why the fixed-cap one-witness relaxation is too strong on an exact six-player boundary: its minimizer lowers overlap only after θ_+ moves from $1/2$ to $1/3$. The seven-player example shows that retaining dispersion optimality still does not prove (CEB). Its positive hinge surplus instead makes the already proved sharpened spectral floor exceed $7/12$. The live route is therefore a spectral-or-common-hinge disjunction at the weaker universal target, not a reciprocal-mass theorem.

The label loss in that disjunction has an exact layer-cake form. Fix the actual ceiling block H, put $R = N \setminus H$ and $m = |R|$, and define $X_t = \{i \in R : x_i \ge t\}$ and $Z_t = \{i \in R : z_i \ge t\}$. If B_H is the spectral expression with H fixed and the residual spectra oppositely paired, then

$$O - B_H = \int_0^\infty \left(|X_t \cap Z_t| - \max\{0, |X_t| + |Z_t| - m\}\right) dt. \tag{LC}$$

Indeed, the first term is the layer-cake formula for residual overlap, while the second is the Fréchet lower intersection bound attained simultaneously by opposite sorting. The integrand is a nonnegative integer. On a fixed pair of ordered payoff partitions, (LC) is therefore a finite nonnegative linear form in the consecutive merged stratum gaps. If B_* denotes the spectral minimum over admissible ceiling assignments, then

$$O = B_* + (B_H - B_*) + (O - B_H),$$

separating top-selection currency from residual label currency. This is a proved identity, not the missing inequality: the remaining common-hinge problem is to show that the two currencies cover $\max\{0, nT/\lfloor n^2/4 \rfloor - B_*\}$. It does, exactly, identify a linear stratum-gap target for the rooted branch-circuit and relative-facet program without conflating those relative circuits with facets of the full exact-game cone.

There is also a direct bridge to the staged transports. If A_t denotes the players in R high in both selections and D_t those low in both, then the integrand in (LC) is exactly $\min\{|A_t|, |D_t|\}$. Indeed $|X_t| + |Z_t| - m = |A_t| - |D_t|$. If player i gains from x to z and a losing carrier j satisfies $x_j \le x_i$, then

$$z_j < x_j \le x_i < z_i,$$

so throughout $(x_j, x_i]$ player i is high in both and j low in both. The mirror carrier condition $z_j \ge z_i$ supplies the interval $(z_i, z_j]$. Thus every sign-coherent staged carrier is a unit rectangle under the label-currency capacity. Any fractional collection whose total occupancy at threshold t is at most $\min\{|A_t|, |D_t|\}$ has total interval length at most $O - B_H$, by integration. The remaining minimal lemma is to pack enough of these carrier intervals to cover the spectral deficit, charging intervals incident with the removed ceiling block to the top-selection currency. This geometric occupancy statement replaces the false raw-coefficient congestion bound.

For fixed payoff strata this packing problem has an exact combinatorial min–max theorem. Split the threshold line into the merged coordinate intervals J_ℓ, let $c_\ell = \min\{|A_t|, |D_t|\}$ there, and let $A_{\ell e}$ indicate whether carrier interval e contains J_ℓ. Then

$$\max\{\textstyle\sum_e |I_e| \rho_e : A\rho \le c, \ \rho \ge 0\} = \min\{\textstyle\sum_\ell c_\ell p_\ell : \sum_{\ell : J_\ell \subset I_e} p_\ell \ge |I_e|, \ p \ge 0\}.$$

The columns of A have the consecutive-ones property, so A is totally unimodular. Consequently an integral-capacity instance has an integral optimal packing with support at most the number of threshold layers, and the primal and dual are exactly computable in polynomial time for rational data. The feasible cover $p_\ell = |J_\ell|$ recovers the label-currency upper bound. Writing $r_k = \sum_{\ell \le k} p_\ell - \beta_k$ turns every carrier-cover constraint for $(\beta_s, \beta_t]$ into $r_t \ge r_s$ and nonnegativity into $r_k - r_{k-1} \ge -|J_k|$. Hence an extreme cover has a rooted spanning-tree certificate in the graph of tight path and carrier arcs. This is a fixed-stratum packing theorem and an exact interface to the rooted incidence packets; it is not by itself a facet theorem for the exact-game cone. The unresolved structural step is to prove that the intervals and supplies obtained from the staged circuits force the packing optimum to pay the post-spectral deficit.

There is a precise linear version of this missing step. Normalize every chosen staged relation at its forced player column and bound each carrier weight by the corresponding auxiliary-player coefficient. Simultaneously choosing the staged relations and the packing is then one rational feasibility problem. In the two hostile controls where the spectral and top-selection terms do not already reach the target, this strengthened system has short exact witnesses. Respectively,

$$\mathbf{1}_{\{0,2,4\}} + \mathbf{1}_{\{0,1,2,3,6\}} + \mathbf{1}_{\{4,6\}} + 2e_5 + e_1 + e_3 = 2 \cdot \mathbf{1}_N$$

supplies $(1/2, 8]$ at weight $1/5$, paying the deficit $3/2$, and

$$\mathbf{1}_{\{1,3,4,5,6\}} + e_2 + e_0 = \mathbf{1}_N$$

supplies $(1/2, 19/2]$ at weight $1/9$, paying the deficit 1. These checks show that the staged-supply formulation survives the known controls; they do not prove its universal feasibility.

The rooted-tree description also gives an exact fixed-dimension decision procedure. Refine the exact-game cone into rational polyhedral cells fixing the active staged bases, payoff orders, ceiling block, spectral branch, and carrier endpoints. On each cell the selections and interval lengths are rational linear in the game, whereas the interval and staged-incidence matrices are fixed. Enumerating rooted cover trees (or bases of the supplied joint system) reduces the existence of a violating game to finitely many exact rational linear programs. This procedure may be exponential or worse and its cellwise inequalities are only candidate full-cone facets: they must still glue across cells and lift to indecomposable semi-balanced systems.

Not claimed. Exactness theory, lower-envelope exactification, and the characterizations of the egalitarian selections are prior art (Arin and Iñarra, 2001; Arin et al., 2003, 2008; Biswas et al., 1999; Dietzenbacher, 2021; Dietzenbacher and Yanovskaya, 2020; Llerena and Mauri, 2017; Csóka et al., 2011; Frank and Murota, 2022; Kroupa and Studený, 2019; Studený and Kratochvíl, 2022; Gong et al., 2022; Gong and Dietzenbacher, 2026); the four-player rigidity is the companion paper's classification (Flanagan, 2026) combined with the Lorenz implication of Llerena and Mauri (2017). The candidate new content is the two-generator core duality and its one-sided floor certificate, the family $G(p, k)$ with its closed forms and deficit identity, the resulting evaluation $\sup_n s(n) = 2$, the identification of the $n = 5$ anomaly with a proof by hand, the one-coalition certificate theorem and the pinned extremal configuration at five players, the evaluation $s(5) = \frac{1}{2}$ with the $(4, 1)$-regime bound $\frac{1}{3}$, and the two closed routes of Section 6, together with the direct exactness- exchange evaluation $s(6) = \frac{2}{3}$. Novelty remains conditional on specialist review.

Reproducibility note

Every theorem not identified below is proved by hand and depends on no program. Theorems 23, 32, 33, and 35 depend on exhaustive finite exact-rational checks, described in their proofs; those checks are proof dependencies, archived in the public repository https://github.com/mfflanag1/counterexample-hunt on branch `codex/fixed-n-handover-facets`, with the theorem-to-driver map in `coopgames/verifiers/README.md` (for Theorems 32 and 33, both pipelines, the coverage audit, and the full control suite; for Theorem 35, the fifteen source-regenerated QF_LRA formulas, Z3/cvc5 records, trust-disabled CPC proofs, and Ethos checks under `s6/strict_exchange_*`, plus the independently formulated bound-two reconstruction under `s6/independent_bound_two/` and the standard-library direct SMT-LIB reconstruction under `s6/third_bound_two/`). The literal 72-assertion human-to-machine map is audited under `s6/strict_exchange_formula_interface.json`; a separately written semantic reconstruction is frozen as `s6/strict_exchange_semantic_interface.json`. These are transcription controls and do not replace external review of the human implication. The orbit anatomy under `s6/strict_exchange_large_case_templates.json` is explanatory; the later 19-orbit formulas under `s6/orbit_core_certificates/` are separately proof-checked reduced alternatives for the three large cases. The anchor-localized $(4, 2, 0)$ alternative under `s6/p4_anchor_core_certificates/` is separately proof-checked and uses only one singleton exactness witness. Its explanatory eleven-row forced-blocker decomposition is regenerated by `s6/audit_strict_exchange_p4_packet_classes.py`; its forced-exchange trees are regenerated by `s6/audit_strict_exchange_p4_relay_trees.py`. Theorem 25 is proved by hand; its superseded finite check and the hand proof's own symbolic guards are archived alongside. The remaining programs are bug audits, not proof dependencies. Exact rational programs in the same repository on branch `codex/hybrid-overlap-facets` archive the refuted effective-bipartition bridge, its exact seven-player counterexample, the dispersion primal/dual witnesses, and the threshold-label currency audit; the latter also replays the exact interval-packing/cover duals and rooted potential trees. A separate exact feasibility replay archives the two displayed staged-supply witnesses; none is a proof dependency for the open all-dimension target. Other programs independently reconstruct all displayed games and coalition values, the closed forms of Theorem 13 at twelve parameter pairs, the full coalition-inequality check and the tight-set prediction of Lemma 14 for every (p, k) with $1 \le p \le 7$, $2 \le k \le 7$ and $n \le 13$, and the four-way agreement on the game of Theorem 21. They are bug audits, not proof dependencies. All arithmetic is exact rational; no floating-point value appears in any stated result. Reported optima come from a from-scratch exact-rational two-phase simplex with Bland's rule, from a staged exact solver over linear real arithmetic, or from exact vertex enumeration, with at least two methods agreeing on every displayed allocation; one library simplex implementation cycles on these degenerate programs and was not used for any reported number.

References

Arin J, Iñarra E (2001) Egalitarian solutions in the core. International Journal of Game Theory 30:187–193. https://doi.org/10.1007/s001820100073

Arin J, Kuipers J, Vermeulen D (2003) Some characterizations of egalitarian solutions on classes of TU-games. Mathematical Social Sciences 46:327–345. https://doi.org/10.1016/S0165-4896(03)00051-9

Arin J, Kuipers J, Vermeulen D (2008) An axiomatic approach to egalitarianism in TU-games. International Journal of Game Theory 37:565–580. https://doi.org/10.1007/s00182-008-0133-6

Biswas AK, Parthasarathy T, Potters JAM, Voorneveld M (1999) Large cores and exactness. Games and Economic Behavior 28:1–12. https://doi.org/10.1006/game.1998.0686

Csóka P, Herings PJJ, Kóczy LÁ (2011) Balancedness conditions for exact games. Mathematical Methods of Operations Research 74:41–52. https://doi.org/10.1007/s00186-011-0353-z

Fujishige S (1980) Lexicographically optimal base of a polymatroid with respect to a weight vector. Mathematics of Operations Research 5:186–196.

Frank A, Murota K (2022) Decreasing minimization on base-polyhedra: relation between discrete and continuous cases. Preprint, arXiv:2202.06561.

Gangam RR, Garg N, Shahkar P, Vazirani VV (2024) Equitable core imputations for max-flow, MST and b-matching games. Preprint, arXiv:2403.06037.

Izquierdo JM, Montes J, Rafels C (2024) Population Lorenz-monotonic allocation schemes for TU-games. Social Choice and Welfare 63:417–436. https://doi.org/10.1007/s00355-024-01538-x

Gong D, Dietzenbacher B (2026) Core bound reduced games and consistency. International Journal of Game Theory 55:13. https://doi.org/10.1007/s00182-026-00986-2

Gong D, Dietzenbacher B, Peters H (2022) Reduced two-bound core games. Mathematical Methods of Operations Research 96:447–457. https://doi.org/10.1007/s00186-022-00800-1

Gong D, Dietzenbacher B, Peters H (2024) One-bound core games. International Journal of Game Theory 53:859–878. https://doi.org/10.1007/s00182-024-00889-0

Klijn F, Vermeulen D, Hamers H, Solymosi T, Tijs S, Villar JP (2003) Neighbor games and the leximax solution. Mathematical Methods of Operations Research 58:191–208.

Kroupa T, Studený M (2019) Facets of the cone of totally balanced games. Mathematical Methods of Operations Research 90:271–300. https://doi.org/10.1007/s00186-019-00672-y

Llerena F, Mauri L (2016) Reduced games and egalitarian solutions. International Journal of Game Theory 45:1053–1069.

Vazirani VV (2024) Fair core imputations for the assignment game: new solution concepts and efficient algorithms. Preprint, arXiv:2402.11437.

Dutta B, Ray D (1989) A concept of egalitarianism under participation constraints. Econometrica 57:615–635.

Dietzenbacher B (2021) Monotonicity and egalitarianism. Games and Economic Behavior 127:194–205. https://doi.org/10.1016/j.geb.2021.03.006

Dietzenbacher B, Yanovskaya E (2020) Antiduality in exact partition games. Mathematical Social Sciences 108:116–121. https://doi.org/10.1016/j.mathsocsci.2020.04.006

Dietzenbacher B, Doğan E (2025) Population monotonicity and egalitarianism. International Journal of Game Theory 54, article 42. https://doi.org/10.1007/s00182-025-00966-y

Flanagan M (2026) Balanced transport at the egalitarian core: a complete boundary phase diagram. Preprint; companion paper.

Llerena F, Mauri L (2017) On the existence of the Dutta–Ray's egalitarian solution. Mathematical Social Sciences 89:92–99. https://doi.org/10.1016/j.mathsocsci.2017.07.002

Studený M, Kratochvíl V (2022) Facets of the cone of exact games. Mathematical Methods of Operations Research 95:35–80. https://doi.org/10.1007/s00186-022-00770-4

Schoot Uiterkamp MHH (2024) A characterization of simultaneous optimization, majorization, and (bi-)submodular polyhedra. Mathematics of Operations Research 50:252–276. https://doi.org/10.1287/moor.2023.0054